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CATHETER ROBOT MODULE FOR TRANSLATION AND ROTATION OF A FLEXIBLE ELONGATED MEDICAL ELEMENT

Abstract

Disclosed is a catheter robot module for translation and rotation of a flexible elongated medical element, including: a casing, two pairs of movable pads: the pads of a same pair at least partly facing each other, each pair of movable pads being adapted to separately or in combination: perform a translation of the flexible elongated medical element longitudinally with respect to the casing, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, perform a rotation of the flexible elongated medical element around longitudinal axis with respect to the casing, by a second rotation cycle which clamps, performs a relative forth translation of the pads in opposite directions, unclamps, performs a relative back translation of the pads in opposite directions, depending on a set rotation direction.

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Background/Summary

[0001] This application is the U.S. national phase of International Application No. PCT/IB2020/001116 filed Dec. 26, 2020, which designated the U.S., the entire contents of which is hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] The invention relates to the technical field of catheter robot modules for translation and rotation of a flexible elongated medical element. This flexible elongated medical element can be guide of a catheter and/or a catheter, and/or a catheter guide. Usually, these elements are disposed so that, at least partly, i.e. on part of their respective length, the catheter guide surrounds the catheter which itself surrounds the guide of a catheter.

BACKGROUND OF THE INVENTION

[0003] According to a prior art, described in EP 15733825, and belonging to the same assignee Robocath, a catheter robot module is described which includes a pair of movable pads configured to clamp and unclamp a flexible elongated medical element. This pair of movable pads is also disposed so as to be able to impart to this flexible elongated medical element, either a translation move and/or a rotation move. This pair of movable pads can translate the flexible elongated medical element like fingers of two hands would pull this flexible elongated medical element forward. This pair of movable pads can rotate the flexible elongated medical element like fingers of a hand would make this flexible elongated medical element roll between those fingers.

[0004] Unlike the manual move of hand fingers, which is first rather slow and which is second assisted by the practitioner's brain whose hands manipulates the flexible elongated medical element, the catheter robot module becomes more and more interesting when the translation and rotation moves of the flexible elongated medical element can be made quicker and quicker. However, the translation speed and rotation speed, as well as the translation speed variations and the rotation speed variations, become rapidly limited, because global synchronization, between on one side translation and rotation moves of each pair of movable pads, and between on the other side both pairs of movable pads, each pair performing translation and rotation, soon becomes hard to manage, when the translation and rotation speeds increase, and also when the rapidity of variations allowed for these translation and rotation speeds increase. Increasing speeds and speed variations, not only improves catheter robot module efficiency to provide the physician with agility in difficult situations such as crossing a lesion or selecting an arterial side branch, but also increases its security allowing for quick reactions in case of incident or in case of danger risk.

[0005] No prior art has tried and tackled these synchronization problems when speeds and speed variations increase. When looking into them, these synchronization problems appear to be at first sight complex and intricate.

SUMMARY OF THE INVENTION

[0006] The object of the present invention is to alleviate at least partly the above-mentioned drawbacks.

[0007] More particularly, the invention technical contribution is of two kinds: [0008] First, it has split and ordered this complex global synchronization task into at least two more simple specific synchronization tasks, which are: [0009] Control of opposition phase between the two pairs of movable pads, [0010] And management of clamping conflict between the two pairs of movable pads, [0011] Second, it has provided for each one of these two specific synchronization tasks: [0012] Not only an efficient technical solution, [0013] But also a technical solution which is compatible with the technical solution of the other specific synchronization task, and can be synchronized with it, [0014] Thereby allowing for use of both these technical solutions to provide a global answer, to the complex global synchronization task existing between both pairs of movable pads.

[0015] So, technical contribution of the invention includes: [0016] Spitting global complex synchronization task between both pairs of movable pads into two specific simpler synchronization tasks which are, first controlling and keeping phase opposition between both pairs of movable pads, and second always keeping at least one pair of movable pads clamped on the flexible elongated medical element, [0017] Bringing technical solutions respectively to these two specific simpler synchronization tasks, which: [0018] Not only solve the problems of these two specific simpler synchronization tasks, [0019] But also solve these problems in a compatible and even more easily synchronized way, so that the problems of both these two specific simpler synchronization tasks can be solved simultaneously, while keeping the global catheter robot module in a reasonable complexity and cost.

[0020] However, the present invention mainly focuses on providing a specific technical solution to the specific simpler synchronization task existing between both pairs of movable pads, which deals with always keeping at least one pair of movable pads clamped on the flexible elongated medical element, what will be useful for improving the process and going toward a rather rapid, fluid and secure control of the moving flexible elongated medical element.

[0021] The main technical contribution of the invention deals with: [0022] A variation of a travel extension of a forth translation in a first translation cycle for at least one of said pairs, and/or variation of a travel extension and/or a duration of a forth translation in a second rotation cycle for at least one of the two pairs of movable pads, [0023] So as to always keep at least one pair of movable pads clamped on the flexible elongated medical element.

[0024] In a preferred embodiment, so as to further improve the process and to get at a rapid, fluid and secure control of the moving flexible elongated medical element, there is a supplementary technical contribution of the invention which deals with: [0025] An additional variation of a duration of a translating back phase in a first translation cycle for at least one of the two pairs of movable pads, [0026] So as to control and keep phase opposition between both pairs of movable pads.

[0027] This object is achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0028] a casing, [0029] two pairs of movable pads: [0030] said pads of a same pair at least partly facing each other, [0031] each pair of movable pads being adapted to, separately or in combination: [0032] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle: [0033] clamping said flexible elongated medical element between said pads, [0034] translating forth said pads synchronously longitudinally in the same direction with respect to said casing, with respect to a user set longitudinal translation direction, [0035] unclamping said flexible elongated medical element, [0036] translating back said pads synchronously longitudinally in the same reverse direction with respect to said casing, with respect to said user set longitudinal translation direction, [0037] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle: [0038] clamping said flexible elongated medical element between said pads, [0039] performing a relative forth translation of said pads transversely in opposite directions with respect to said casing, with respect to a transversal translation direction corresponding to a set rotation direction, [0040] unclamping said flexible elongated medical element, [0041] performing a relative back translation of said pads transversely in opposite reverse directions with respect to said casing, with respect to said transversal translation direction corresponding to said set rotation direction, [0042] a driver of said pairs of movable pads implemented so that: [0043] in at least one mode where, in combination, said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, and said rotation of said flexible elongated medical element is performed by at least one of said pairs of movable pads, [0044] conflict of synchronization between said translation and said rotation is managed at least: [0045] by varying travel extension of said forth translation in said first translation cycle for at least one of said pairs, and/or by varying travel extension and/or duration of said forth translation in said second rotation cycle for at least one of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first translation cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second rotation cycle.

[0046] Preferably, said driver of said pairs of movable pads is also implemented so that: [0047] in one or several or all modes where said translation of said flexible elongated medical element is performed: [0048] said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled at least: [0049] by varying duration of said translating back in said first translation cycle for at least one of said pairs, [0050] so as to control and keep said phase opposition between both said pairs.

[0051] This object can be achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0052] a casing, [0053] two pairs of movable pads: [0054] said pads of a same pair at least partly facing each other, [0055] each pair of movable pads being adapted to, separately or in combination: [0056] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle: [0057] clamping said flexible elongated medical element between said pads, [0058] translating forth said pads synchronously longitudinally in the same direction with respect to said casing, with respect to a user set longitudinal translation direction, [0059] unclamping said flexible elongated medical element, [0060] translating back said pads synchronously longitudinally in the same reverse direction with respect to said casing, with respect to said user set longitudinal translation direction, [0061] like fingers of a hand pulling said flexible elongated medical element forward, [0062] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle: [0063] clamping said flexible elongated medical element between said pads, [0064] performing a relative forth translation of said pads transversely in opposite directions with respect to said casing, with respect to a transversal translation direction corresponding to a set rotation direction, [0065] unclamping said flexible elongated medical element, [0066] performing a relative back translation of said pads transversely in opposite reverse directions with respect to said casing, with respect to said transversal translation direction corresponding to said set rotation direction, [0067] like fingers of a hand making said flexible elongated medical element rolling between them, [0068] a driver of said pairs of movable pads implemented so that: [0069] in at least one mode where, in combination, said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, and said rotation of said flexible elongated medical element is performed by at least one of said pairs of movable pads, [0070] conflict of synchronization between said translation and said rotation is managed at least: [0071] by varying travel extension of said forth translation in said first translation cycle for at least one of said pairs, and/or by varying travel extension and/or duration of said forth translation in said second rotation cycle for at least one of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first translation cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second rotation cycle.

[0072] Preferably, said driver of said pairs of movable pads is also implemented so that: [0073] in one or several or all modes where said translation of said flexible elongated medical element is performed: [0074] said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled at least: [0075] by varying duration of said translating back in said first translation cycle for at least one of said pairs, [0076] so as to control and keep said phase

opposition between both said pairs.
[0077] This object is also achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0078] a casing, [0079] two pairs of movable pads: [0080] said pads of a same pair at least partly facing each other, [0081] each pair of movable pads being adapted to, separately or in combination: [0082] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, [0083] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle which clamps, performs a relative forth translation of said pads in opposite directions, unclamps, performs a relative back translation of said pads in opposite directions, depending on a set rotation direction, [0084] a driver of said pairs of movable pads implemented so that: [0085] conflict of synchronization, between said translation alternatively performed by said pairs of movable pads working in phase opposition and said rotation, when in combination, is managed at least: [0086] by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for at least one of said pairs, [0087] so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during said translation and during said rotation.
[0088] Preferably, said driver of said pairs of movable pads is also implemented so that: [0089] said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled at least: [0090] by varying duration of said translating back in said first translation cycle for at least one of said pairs, [0091] so as to control and keep said phase opposition between both said pairs.
[0092] This object can be also achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0093] a casing, [0094] two pairs of movable pads: [0095] said pads of a same pair at least partly facing each other, [0096] each pair of movable pads being adapted to, separately or in combination: [0097] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, like fingers of a hand pulling said flexible elongated medical element forward, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, [0098] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, like fingers of a hand making said flexible elongated medical element rolling between them, by a second rotation cycle which clamps, performs a relative forth translation of said pads in opposite directions, unclamps, performs a relative back translation of said pads in opposite directions, depending on a set rotation direction, [0099] a driver of said pairs of movable pads implemented so that: [0100] conflict of synchronization, between said translation alternatively performed by said pairs of movable pads working in phase opposition and said rotation, when in combination, is managed at least: [0101] by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for at least one of said pairs, [0102] so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during said translation and during said rotation.
[0103] Preferably, said driver of said pairs of movable pads is also implemented so that: [0104] said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled at least: [0105] by varying duration of said translating back in said first translation cycle for at least one of said pairs, [0106] so as to control and keep said phase opposition between both said pairs.
[0107] Said rotation of said flexible elongated medical element may be performed by only one of said pairs of movable pads.

[0108] Preferably, said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads. Hence, global move of the flexible elongated medical element can be made more fluid and rapid, to the cost of an additional complexity, the ability of a second pair of movable pads to perform rotation of the flexible elongated element.
[0109] In this later case, said rotation of said flexible elongated medical element is preferably alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled, at least by varying duration of said translating back in said second rotation cycle for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.
[0110] In the case where said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, this object is still achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0111] a casing, [0112] two pairs of movable pads: [0113] said pads of a same pair at least partly facing each other, [0114] each pair of movable pads being adapted to separately or in combination: [0115] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle: [0116] clamping said flexible elongated medical element between said pads, [0117] translating forth said pads synchronously longitudinally in the same direction with respect to said casing, with respect to a user set longitudinal translation direction, [0118] unclamping said flexible elongated medical element, [0119] translating back said pads synchronously longitudinally in the same reverse direction with respect to said casing, with respect to said user set longitudinal translation direction, [0120] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle: [0121] clamping said flexible elongated medical element between said pads, [0122] performing a relative forth translation of said pads transversely in opposite directions with respect to said casing, with respect to a transversal translation direction corresponding to a set rotation direction, [0123] unclamping said flexible elongated medical element, [0124] performing a relative back translation of said pads transversely in opposite reverse directions with respect to said casing, with respect to said transversal translation direction corresponding to said set rotation direction, [0125] a driver of said pairs of movable pads implemented so that: [0126] said translation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, [0127] said rotation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, [0128] conflict of synchronization between said translation and said rotation being managed, [0129] at least: [0130] by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for both of said pairs, [0131] so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first translation cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second rotation cycle.
[0132] Preferably, said driver of said pairs of movable pads is also implemented so that: [0133] said phase opposition is controlled, at least: [0134] by varying duration of said translating back, in said first translation cycle and in said second rotation cycle, for at least one of said pairs, [0135] so as to control and keep said phase opposition between both said pairs.
[0136] In the case where said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, this object can be still achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0137] a casing, [0138] two pairs of movable pads: [0139] said pads of a same pair at least partly facing each other, [0140] each pair of movable pads being adapted to separately or in combination: [0141] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle: [0142] clamping said flexible elongated medical element

between said pads, [0143] translating forth said pads synchronously longitudinally in the same direction with respect to said casing, with respect to a user set longitudinal translation direction, [0144] unclamping said flexible elongated medical element, [0145] translating back said pads synchronously longitudinally in the same reverse direction with respect to said casing, with respect to said user set longitudinal translation direction, [0146] like fingers of a hand pulling said flexible elongated medical element forward, [0147] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle: [0148] clamping said flexible elongated medical element between said pads, [0149] performing a relative forth translation of said pads transversely in opposite directions with respect to said casing, with respect to a transversal translation direction corresponding to a set rotation direction, [0150] unclamping said flexible elongated medical element, [0151] performing a relative back translation of said pads transversely in opposite reverse directions with respect to said casing, with respect to said transversal translation direction corresponding to said set rotation direction, [0152] like fingers of a hand making said flexible elongated medical element rolling between them, [0153] a driver of said pairs of movable pads implemented so that: [0154] said translation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, [0155] said rotation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, p**2** conflict of synchronization between said translation and said rotation being managed, [0156] at least: [0157] by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for both of said pairs, [0158] so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first translation cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second rotation cycle.
[0159] Preferably, said driver of said pairs of movable pads is also implemented so that: [0160] said phase opposition is controlled, at least: [0161] by varying duration of said translating back, in said first translation cycle and in said second rotation cycle, for at least one of said pairs, [0162] so as to control and keep said phase opposition between both said pairs.
[0163] In the case where said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, this object is still achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0164] a casing, [0165] two pairs of movable pads: [0166] said pads of a same pair at least partly facing each other, [0167] each pair of movable pads being adapted to separately or in combination: [0168] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, [0169] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle which clamps, performs a relative forth translation of said pads in opposite directions, unclamps, performs a relative back translation of said pads in opposite directions, depending on a set rotation direction, [0170] a driver of said pairs of movable pads implemented so that: [0171] said translation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, [0172] said rotation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, [0173] conflict of synchronization between said translation and said rotation being managed, at least: [0174] by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for both of said pairs, [0175] so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during said translation and during said rotation.
[0176] Preferably, said driver of said pairs of movable pads is also implemented so that: [0177] said phase opposition is controlled, at least: [0178] by varying duration of said translating back, in

said first translation cycle and in said second rotation cycle, for at least one of said pairs, [0179] so as to control and keep said phase opposition between both said pairs.
[0180] In the case where said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, this object can be still achieved with a catheter robot module for translation and rotation of a flexible elongated medical element, comprising: [0181] a casing, [0182] two pairs of movable pads: [0183] said pads of a same pair at least partly facing each other, [0184] each pair of movable pads being adapted to separately or in combination: [0185] perform a translation of said flexible elongated medical element longitudinally with respect to said casing, like fingers of a hand pulling said flexible elongated medical element forward, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, [0186] perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, like fingers of a hand making said flexible elongated medical element rolling between them, by a second rotation cycle which clamps, performs a relative forth translation of said pads in opposite directions, unclamps, performs a relative back translation of said pads in opposite directions, depending on a set rotation direction, [0187] a driver of said pairs of movable pads implemented so that: [0188] said translation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, [0189] said rotation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, [0190] conflict of synchronization between said translation and said rotation being managed, at least: [0191] by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for both of said pairs, [0192] so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during said translation and during said rotation.
[0193] Preferably, said driver of said pairs of movable pads is also implemented so that: [0194] said phase opposition is controlled, at least: [0195] by varying duration of said translating back, in said first translation cycle and in said second rotation cycle, for at least one of said pairs, [0196] so as to control and keep said phase opposition between both said pairs.
[0197] Preferred embodiments comprise one or more of the following features, which can be taken separately or together, either in partial combination or in full combination.
[0198] Preferably, said driver of said pairs of movable pads is implemented so that said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both pairs working in phase opposition, said phase opposition being controlled mainly or only, by varying duration of said translating back, in said first translation cycle for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.
[0199] Hence, variation of this key parameter, duration of said translating back, in said first translation cycle for at least one of said pairs, may be sufficient to control and keep said phase opposition between both said pairs.
[0200] Preferably, said driver of said pairs of movable pads is implemented so that said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, conflict of synchronization between said translation and said rotation is managed mainly or only, by varying travel extension of said forth translation, in said first translation cycle for at least one of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second cycle.
[0201] Hence, variation of this key parameter, travel extension of said forth translation, in said first translation cycle for at least one of said pairs, may be sufficient to always keep at least one pair of movable pads clamped on said flexible elongated medical element.
[0202] Preferably, said forth translation duration is always longer than said back translation

duration.

[0203] Hence, the catheter robot module is more efficient since main part of the time is dedicated to move the flexible elongated medical element in the targeted direction rather than to bring back the pairs of movable pads in the reverse direction for next pull of this flexible elongated medical element.

[0204] Preferably, said varying travel extension of said forth translation in said first translation cycle for one of said pairs is performed by extending a predetermined standard forth translation travel range, reaching a value ranging from said predetermined standard forth translation travel range to a predetermined maximum forth translation travel range.

[0205] Hence, this is a simple and efficient way to provide for a delay before unclamping, by providing for an extra travel in such a way that the perturbation before getting back at standard path with synchronized moves between the two pairs of movable pads is minimized.

[0206] Preferably, said predetermined maximum forth translation travel range is comprised between 110% and 150% of said predetermined standard forth translation travel range, preferably between 120% and 140% of said predetermined standard forth translation travel range.

[0207] Hence, there is a good compromise between: [0208] the global efficiency in most cases where only the standard travel is needed, [0209] and the high security in few cases where a notable extra travel may be needed.

[0210] Preferably, said predetermined maximum forth translation travel range is split in two equal parts respectively at both ends of said predetermined standard forth translation travel range.

[0211] Hence, both positive and negative targeted speed values chosen by user can be both efficiently managed.

[0212] Preferably, there is some temporary overlapping between said flexible elongated medical element clamping by one of said pairs of movables pads and said flexible elongated medical element clamping by the other one of said pairs of movables pads, said temporary overlapping lasting preferably between 10% and 95% of the whole duration of said translation of said flexible elongated medical element.

[0213] Hence, the security is improved, by increasing the time when both pairs of movable clamps are simultaneously clamped.

[0214] Preferably, said flexible elongated medical element unclamping is performed simultaneously to a portion of said forth translation travel extension, during the second half of said forth translation travel extension, said portion ranging preferably from 5% to 20% of the full extent of said forth translation travel extension.

[0215] Hence, the fluidity of the flexible elongated medical element clamping and bringing back pairs of movable pads before next flexible elongated medical element pulling is improved.

[0216] Preferably, said flexible elongated medical element clamping is performed simultaneously to a portion of said forth translation travel extension, during the first half of said forth translation travel extension, said portion ranging preferably from 5% to 20% of the full extent of said forth translation travel extension.

[0217] Hence, the fluidity of the flexible elongated medical element clamping and pulling is improved.

[0218] Preferably, said flexible elongated medical element clamping starts after the end of said back translation travel extension and after the beginning of next said forth translation travel extension.

[0219] Hence, the fluidity of the flexible elongated medical element clamping and pulling is improved.

[0220] Preferably, said varying duration of said translating back in said first translation cycle for one of said pairs, so as to control and keep said phase opposition between both said pairs, is performed by reducing or extending duration (and thus speed) with respect to a standard back translation duration.

[0221] Hence, this is a simple and efficient way to provide for a resynchronization, by providing for an extra duration range, either to increase or to decrease a standard duration, in such a way that the perturbation before getting back at standard path with synchronized moves between the two pairs of movable pads is minimized.
[0222] Preferably, said varying duration of said translating back in said first translation cycle for one of said pairs, so as to control and keep said phase opposition between both said pairs, is performed by reducing or extending duration with respect to a standard back translation duration less than requested for optimal phase opposition controlling and keeping so as to improve stability to the cost of higher number of cycles to get back at phase opposition target, a factor a of correction attenuation comprised between 0 and 1 being applied.
[0223] Hence, by reducing or extending duration by an amount less than what would be needed for a complete and full phase opposition control, a gain in stability can be obtained, even if such stability is not regained immediately from last instability.
[0224] Preferably, said factor a of correction attenuation is comprised between 0.3 and 0.7, and is preferably about 0.5.
[0225] Hence, the compromise between stability level and rapidity to regain stability after last instability can be optimized.
[0226] Preferably, said standard back translation duration is a decreasing function of a user command speed target value(s), for either translation and/or rotation, preferably minimum of both speed target values, when applicable, becoming selected user command speed target value.
[0227] Hence, the rapidity to correct deviations from standard working is better adapted to the translation and rotation speeds requested by the user.
[0228] Preferably, said decreasing function presents a central curved part which presents a concavity toward top and which is located between two horizontal parts.
[0229] Hence, the two horizontal parts allow for fluid and correct working of the correction. Indeed, the upper horizontal part avoids too long unclamping period which would increase the number of clamping conflicts happening. Indeed, the lower horizontal part avoids too much demand on the response time of the actuators which are limited in speed.
[0230] Preferably, said central curved part is inversely proportional to said selected user command speed target value, whereas said horizontal parts are constant with respect to said selected user command speed target value.
[0231] Hence, the rapidity to correct deviations from standard working is better adapted to the translation and rotation speeds requested by the user.
[0232] Preferably, said user set longitudinal translation direction can be varied continuously by said user, and/or said user set rotation direction can be varied continuously by said user.
[0233] Hence, the catheter robot module is more flexible and thereby more useful to the user.
[0234] Preferably, said translation of said flexible elongated medical element longitudinally for first of said pairs of pads is performed using several steps controlled by a first finite state machine, said rotation of said flexible elongated medical element around longitudinal axis with respect to said casing for first of said pairs of pads is performed using several steps controlled by a second finite state machine, said translation of said flexible elongated medical element longitudinally for second of said pairs of pads is performed using several steps controlled by a third finite state machine, said rotation of said flexible elongated medical element around longitudinal axis with respect to said casing for second of said pairs of pads is performed using several steps controlled by a fourth finite state machine.
[0235] Hence, the compromise between complexity and efficiency is improved.
[0236] Preferably, each of said finite state machines determines for a transition period between said forth translation and said back translation to go progressively from said forth translation to said back translation: start of said transition period, duration of said transition period, end of said transition period.

[0237] Hence, the global process moving the flexible elongated medical element is more fluid.
[0238] Preferably, each of said finite state machines has all its state variables updated periodically with a period which is less than 5 ms, preferably comprised between 0.5 ms and 2 ms, more preferably about 1 ms.
[0239] Hence, the global process moving the flexible elongated medical element is also more reactive, while not losing its fluidity.
[0240] Further features and advantages of the invention will appear from the following description of embodiments of the invention, given as non-limiting examples, with reference to the accompanying drawings listed hereunder.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0241] FIG. **1** shows schematically two pairs of two movable pads each, within an example of a catheter robot module according to the invention.
[0242] FIG. **2** shows schematically the motions of practitioner hands reproduced by an example of a catheter robot module according to the invention.
[0243] FIG. **3** shows schematically the correspondence between practitioner hands and the pairs of movable pads of an example of a catheter robot module according to the invention.
[0244] FIG. **4** shows schematically the different phases of the motions of two pads belonging to a pair of movable pads in an example of a catheter robot module according to the invention.
[0245] FIG. **5** shows schematically an example of a shape of a graph giving a targeted position of a pair of movable pads along a longitudinal axis x as a function of time, the flexible elongated medical element moving along this longitudinal axis x, in an example of a catheter robot module according to the invention.
[0246] FIG. **6** shows schematically an example of a correspondence between a shape of a graph giving a targeted position of a pair of movable pads along a longitudinal axis x as a function of time, the flexible elongated medical element moving along this longitudinal axis x, and a shape of a graph giving an actual position of a pair of movable pads along a longitudinal axis x as a function of time, in an example of a catheter robot module according to the invention.
[0247] FIG. **7** shows schematically an example of a shape of a graph giving a more realistic targeted position of a pair of movable pads along a longitudinal axis x as a function of time, the flexible elongated medical element moving along this longitudinal axis x, in an example of a catheter robot module according to the invention.
[0248] FIG. **8** shows schematically the evolution of clamping curve of the flexible elongated medical element between two movable pads of one of the pairs of movable pads, as a function of time, in an example of a catheter robot module according to the invention.
[0249] FIG. **9** shows schematically the moves of the two pads of a pair of movable pads, within the horizontal plane, in an example of a catheter robot module according to the invention.
[0250] FIG. **10** shows schematically the different phases of a rotation of a flexible elongated medical element between two pads of a pair of movable pads, in an example of a catheter robot module according to the invention.
[0251] FIG. **11** shows schematically graphs of evolution, as a function of time, of clamping states of the two pairs of movables pads, for a translation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.
[0252] FIG. **12** shows schematically graphs of evolution, as a function of time, of clamping states of the two pairs of movables pads, for a slow rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.
[0253] FIG. **13** shows schematically graphs of evolution, as a function of time, of clamping states

of the two pairs of movables pads, for a fast translation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0254] FIG. **14** shows schematically graphs of evolution, as a function of time, of clamping state of one of the pairs of movables pads, for a combined translation and rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0255] FIG. **15** shows schematically graphs of evolution, as a function of time, of clamping state of the other one of the pairs of movables pads, for a combined translation and rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0256] FIG. **16** shows schematically graphs of evolution, as a function of time, of clamping state of both pairs of movables pads, for a combined translation and rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0257] FIG. **17** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a sudden change of user speed setpoint, in an example of a catheter robot module according to the invention.

[0258] FIG. **18** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a change of user speed setpoint, with a desynchronization problem with a fixed duration of U-turn, in an example of a catheter robot module according to the invention.

[0259] FIG. **19** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with a first margin, in an example of a catheter robot module according to the invention.

[0260] FIG. **20** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with use of the first margin, with a positive user setpoint, in an example of a catheter robot module according to the invention.

[0261] FIG. **21** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with use of the first margin, with a negative user setpoint, in an example of a catheter robot module according to the invention.

[0262] FIG. **22** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with use of a second margin, with a positive user setpoint, in an example of a catheter robot module according to the invention.

[0263] FIG. **23** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with synchronized and unsynchronized moves, in an example of a catheter robot module according to the invention.

[0264] FIG. **24** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a first step of correction of desynchronization, in an example of a catheter robot module according to the invention.

[0265] FIG. **25** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a second step of correction of desynchronization, in an example of a catheter robot module according to the invention.

[0266] FIG. **26** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a third step of correction of desynchronization, in an example of a catheter robot module according to the invention.

[0267] FIG. **27** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a fourth step of correction of desynchronization, in an example of a catheter robot module according to the invention.

[0268] FIG. **28** shows schematically a graph of evolution, as a function of user speed setpoint, of a wished U-turn duration, in an example of a catheter robot module according to the invention.

[0269] FIG. **29** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a step of management of clamping conflict, in an example of a catheter robot module according to the invention.

[0270] FIG. **30** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with another step of management of clamping conflict, in an example of a catheter robot module according to the invention.
[0271] FIG. **31** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a progressive change of user speed setpoint, in an example of a catheter robot module according to the invention.
[0272] FIG. **32** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a change of user speed setpoint during the U-turn, in an example of a catheter robot module according to the invention.
[0273] FIG. **33** shows schematically the temporal evolution of four finite state machines, in an example of a catheter robot module according to the invention.
[0274] FIG. **34** shows schematically the 12 states of a cycle of the four finite state machines, in an example of a catheter robot module according to the invention.
[0275] FIG. **35** shows schematically a synoptic representing a cycle of the four finite state machines, in an example of a catheter robot module according to the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0276] The present invention deals with a catheter robot module implementing a process used to coordinate the motion of the actuators in a robotic module designed to move a guidewire or any flexible elongate medical element. It is intended to be part of a robot manipulating flexible elongated medical elements (guidewires, balloon or stent catheters, guiding catheters, etc.) for vascular interventions in various domains (interventional cardiology, interventional neuroradiology, peripheral vascular interventions, etc. . . .). Such a catheter robot module is described in more detail in WO2015189531, which is hereby incorporated by reference.
[0277] Embodiments of the invention relate to a process to control the motions of a robotic platform designed for the manipulation of at least one flexible elongated medical element in the vascular field, said robotic platform containing: [0278] A control unit allowing a user to set the translation and/or rotation speed setpoints of the at least one flexible elongated medical element, in a continuous way, [0279] A robot, which can communicate with the control unit to receive continuously in real time said translation and/or rotation speed setpoints, and which contains at least one robotic module, said robotic module containing: [0280] At least two pairs of fingers manipulating the at least one flexible elongated medical element: [0281] Each finger being able to move along the x, y and z axis, [0282] Each pair of fingers having linked movement along each axis: [0283] X axis (control of translation): identical movements, [0284] Y axis (control of clamping): opposite movements in disjointed range, the distance between the maximum position of the finger with the lower range and the minimum position of the finger with the higher range being approximately equal to the diameter of the flexible elongated medical element, [0285] Z axis (control of rotation): opposite movements in the same range, i.e. when one finger is at the maximum position, the other finger is at the minimum position and vice-versa, to move said flexible elongated medical element in translation and/or rotation according to said translation and/or rotation speed setpoints defined by the user, [0286] Motors, [0287] Mechanical interfaces between the motors and the at least two pairs of fingers, [0288] A sterile interface placed between the at least two pairs of fingers and the at least one flexible elongated medical element, [0289] Electronics and embedded software controlling the motors, [0290] Four Finite State Machines (FSM) to control the motion of the at least two pairs of fingers, through the control of the motors and thanks to the motion transmission of the mechanical interfaces: [0291] The first FSM controlling the motion of the first pair of fingers along the x and y axis (translation of the flexible elongated medical element), [0292] The second FSM controlling the motion of the first pair of fingers along the z and y axis (rotation of the flexible elongated medical element), [0293] The third FSM controlling the motion of the second pair of fingers along the x and y axis (translation of the flexible elongated medical element), [0294] The fourth FSM controlling the motion of the second

pair of fingers along the z and y axis (rotation of the flexible elongated medical element), [0295] Each FSM: [0296] has at least the **4** following phases: [0297] 1. Clamping phase: Clamping the flexible elongated medical element, [0298] 2. Active phase: Moving the flexible elongated medical element in translation (first and third FSM) or rotation (second or fourth FSM) according to the user defined translation and/or rotation speed setpoints, [0299] 3. Unclamping phase: Unclamping the flexible elongated medical element, [0300] 4. U-turn phase: While remaining unclamped, going back to (or near) the initial position of phase **1**, [0301] has a "margin mechanism", activated during the phase **1**, using a "standard" travel range and an "extended" travel range, where a pair of finger uses normally the "standard" range, but continues on the "extended" range in case the other hand is temporarily not clamping the flexible elongated medical element, in order to ensure that at least one of the two pairs of fingers is always clamped, [0302] has a "U-turn duration adaptation mechanism", activated during phase **4**, based on adaptation of the duration of said "U-turn phase", to ensure that the motion along the x axis (for first and third FSM) or z axis (for second and fourth FSM) of the pair of fingers is maintained in phase opposition with the same motion of the other pair, so that there is an optimized cooperation between the two pairs, [0303] where the potential conflict due to both the first and second FSM controlling the motion of the first pair of fingers along the y axis and of both the third and fourth FSM controlling the motion of the second pair of fingers along the y axis is solved in the following way: If one FSM asks for clamping and the other asks for unclamping, then unclamp.
[0304] FIG. **1** shows schematically two pairs of two movable pads each, within an example of a catheter robot module according to the invention.
[0305] A robotic module is composed of **4** pads **11**, **12**, **13** and **14**, each of them being able to move in 3 directions (x, y, z). there are a first pair **15** of pads **11** and **12**, and a second pair **16** of pads **13** and **14**. Pads **11** to **14** first clamp a flexible elongated medical element **10**, and second translate and/or rotate this flexible elongated medical element **10**.
[0306] FIG. **2** shows schematically the motions of practitioner hands reproduced by an example of a catheter robot module according to the invention.
[0307] These pads **11** to **14** are equivalent to **4** fingers **21**, **22**, **23** and **24**, manipulating a tube **10** as shown on FIG. **2**. The tube is translated and rotated as would do fingers **21** and **22** of left hand **25** as well as fingers **23** and **24** of right hand **26**.
[0308] FIG. **3** shows schematically the correspondence between practitioner hands and the pairs of movable pads of an example of a catheter robot module according to the invention.
[0309] As in the FIG. **2**, there are two pairs **15** and **16**, respectively of pads **11** and **12**, and **13** and **14**. These two pairs **15** and **16** will be called them the "left hand" **15** and the "right hand" **16**. More generally speaking, "hand" will refer to a pair of pads.
[0310] FIG. **4** shows schematically the different phases of the motions of two pads belonging to a pair of movable pads in an example of a catheter robot module according to the invention.
[0311] The translation motion is obtained thanks to combined motions of the pads in the x and y directions. The FIG. **4** illustrates this motion on one hand, what means on one pair of pads.
[0312] The following steps are repeated: [0313] Phase a: clamping of pads **41** and **42**, (by an y axis motion, [0314] Phase b: translation of pads **43** and **44**, by an x axis motion, [0315] Phase c: unclamping of pads **45** and **46**, by an y axis motion, [0316] Phase d: go back to initial position or "U-turn", of pads **47** and **48**, by an x axis motion.
[0317] FIG. **5** shows schematically an example of a shape of a graph giving a targeted position of a pair of movable pads along a longitudinal axis x as a function of time, the flexible elongated medical element moving along this longitudinal axis x, in an example of a catheter robot module according to the invention.
[0318] During the "active" phase **51**, the pads **11** and **12** for instance, and/or pads **13** and **14** as may be the case, are clamped and the speed of these pads along the x axis corresponds to the wanted flexible elongated medical element speed, as defined by the user from the control unit, using for

example using joysticks. This corresponds to the phase b of the FIG. **4**.

[0319] During the "U-turn" phase **52**, the pads **11** and **12** are unclamped, and they travel along the x axis in the opposite direction to go back to their initial position and be ready for the next active phase **51**. This corresponds to the phases c, d and a of the FIG. **4**.

[0320] Here, a cycle with two phases is described, an active and a U-turn phase. This is a simplified cycle for the sake of clarity. This will be refined below, because a cycle actually breaks down into more than two phases.

[0321] In this example, we have represented a U-turn phase **52** with a linear path. In practice no physical system could follow such a path, because sudden speed changes would imply infinite accelerations.

[0322] FIG. **6** shows schematically an example of a correspondence between a shape of a graph giving a targeted position of a pair of movable pads along a longitudinal axis x as a function of time, the flexible elongated medical element moving along this longitudinal axis x, and a shape of a graph giving an actual position of a pair of movable pads along a longitudinal axis x as a function of time, in an example of a catheter robot module according to the invention.

[0323] The FIG. **6**text missing or illegible when filed shows the actual position of the pads **11** and **12**, with active phase **63** and U-turn phase **62**, both active phase **63** and U-turn phase **62** being linked by a rounded junction **64**. This is to be compared with the theoretical triangular path **61**.

[0324] FIG. **7** shows schematically an example of a shape of a graph giving a more realistic targeted position of a pair of movable pads along a longitudinal axis x as a function of time, the flexible elongated medical element moving along this longitudinal axis x, in an example of a catheter robot module according to the invention.

[0325] For sake of clarity, the embodiment of the invention will be described with a target path of the actuators which are triangular curves. Other embodiments are possible, and may be better, where the U-turn phase has a more "rounded" shape, as illustrated in FIG. **7**. FIG. **7** shows active phase **71** followed by U-turn phase **72**, both being linked together by a rounded part **74**. Such a curve will result in less effort requirements for the actuators as well as less stress for the mechanics.

[0326] FIG. **8** shows schematically the evolution of clamping curve of the flexible elongated medical element between two movable pads of one of the pairs of movable pads, as a function of time, in an example of a catheter robot module according to the invention.

[0327] To implement the motion of the pads **11** and **12** aiming at translating a flexible elongated medical element along the x axis, the pads also need to move along the y axis to clamp and unclamp the flexible elongated medical element. These motions along two different axes have to be synchronized as illustrated in FIG. **8**, to produce the wanted motion of the pads **11** and **12**. On FIG. **8**, the motion of the two pads, (first Pad **1** & second Pad **2**) of one hand, may be seen, either pads **11** and **12** of pair **15**, or pads **13** and **14** of pair **16**. In a cycle, there are successively following phases: active phase **81**, unclamping phase **82**, U-turn phase **83**, clamping phase **84**, and again active phase **81** of next cycle.

[0328] The two pads **11** and **12** get closer during the clamping phase **84**. They do not touch each other, though, because the flexible elongated medical element is between them: their closest distance corresponds to the flexible elongated medical element diameter, noted "device diameter" on FIG. **8**.

[0329] The cycle of a pair **15** includes the following phases: active phase **85** translating the flexible elongated medical element with both pads **11** and **12** clamped around this flexible elongated medical element, then unclamping phase **86** with both pads **11** and **12** releasing this flexible elongated medical element, then U-turn phase **87** with pads **11** and **12** getting back towards their initial position, then clamping phase **88** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element, and then again active phase **85** of next cycle.

[0330] FIG. **9** shows schematically the moves of the two pads of a pair of movable pads, within the horizontal plane, in an example of a catheter robot module according to the invention.

[0331] The cycle of a pair **15** of pads **11** and **12** includes the following phases: active phase **91** translating the flexible elongated medical element with both pads **11** and **12** clamped around this flexible elongated medical element **10**, then unclamping phase **92** with both pads **11** and **12** releasing this flexible elongated medical element, then U-turn phase **93** with pads **11** and **12** getting back towards their initial position, then clamping phase **94** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **91** of next cycle.
[0332] The FIG. **8** shows that re-clamping after the U-turn phase **87** starts with a small delay after the phase **83**. This is a wanted effect in order to take into account the response time of the actuators as explained on FIG. **6**. This results in a small "appendix" **95** on the left of the pads path, as shown on FIG. **9** text missing or illegible when filed.
[0333] The maximum distance between the two pads **11** and **12** along the y axis during unclamping has been voluntarily exaggerated for a better readability. In practice, the unclamping distance should be minimized in order to reduce power consumption and increase performance.
[0334] FIG. **10** shows schematically the different phases of a rotation of a flexible elongated medical element between two pads of a pair of movable pads, in an example of a catheter robot module according to the invention.
[0335] Rotation is based on the same principle as translation, except that the motion uses the RR (right rotation) and LR (left rotation) actuators instead of RT (right translation) and LT (left translation). The flexible elongated medical element rolls between the two pads of one hand, as illustrated in FIG. **10** text missing or illegible when filed.
[0336] A rotating cycle has following phases, from left to right sides of FIG. **10**: [0337] First, pads **11** and **12** of pair **15** clamp flexible elongated medical element **10**, [0338] Then, pads **11** and **12** of pair **15** translate in opposite directions to make flexible elongated medical element **10** rotate between them, like fingers making a tube or a cigarette rolling between them, [0339] Then, pads **11** and **12** of pair **15** unclamp flexible elongated medical element **10**, [0340] Then, pads **11** and **12** of pair **15** go back to their initial positions with respect to flexible elongated medical element **10**, [0341] Then, pads **11** and **12** of pair **15** clamp flexible elongated medical element **10**, starting thereby a new rotation cycle.
[0342] Since translation and rotation motions are independent, it is possible to combine them to translate and rotate the flexible elongated medical element **10** simultaneously. The translation uses the actuators along the x axis, while simultaneously the actuators along the z axis enable the rotation.
[0343] Each hand has therefore two pads, pads **11** and **12** for pair **15** (left hand), and pads **13** and **14** for pair **16** (right hand), each of them having 3 degrees of freedom. This seems at first sight to result in 6 degrees of freedom per hand, and a total of 12 degrees of freedom.
[0344] However, some motions are linked, thus reducing the total number of degrees of freedom: . [0345] On the x axis, both pads **11** and **12** of the same hand have the same motion,. [0346] On the z axis, both pads **11** and **12** of the same hand have opposite motions: in order to roll the flexible elongated medical element **10** between the pads **11** and **12**, one goes up while the other goes down, thereby making the flexible elongated medical element **10** rolling between the pads **11** and **12**, [0347] On the y axis, both pads **11** and **12** of the same hand have opposite motions: in order to clamp, they have to go closer to each another, and, to unclamp, they need to move away from each other.
[0348] Since there is a coupling between the two pads on each axis, this reduces the number of degrees of freedom to 6.
[0349] The actuators will be named, by using the following abbreviations, as mentioned in table 1. TABLE-US-00001 TABLE 1 Used in Used in flexible elongated flexible elongated Abbre- medical element medical element viation Meaning Axis translation rotation RT Right Translation X ✓ RR Right Rotation Z ✓ RC Right Clamping Y ✓ ✓ LT Left Translation X ✓ LR Left Rotation Z ✓ LC

Left Clamping	Y	✓	✓

[0350] This table 1 has 6 lines, corresponding to the **6** degrees of freedom listed above. Each line therefore represents the motion of the two pads **11** and **12** (or **13** and **14**) of one hand (corresponding to pair **15** or to pair **16**) along one axis.
[0351] Since there is a fixed link between the motion of the two pads **11** and **12** of the same hand, there is no more need to show the motion of both pads of both hands, as on FIG. **8**, but showing motion of one pad of each hand will be considered as sufficient.
[0352] FIG. **11** shows schematically graphs of evolution, as a function of time, of clamping states of the two pairs of movables pads, for a translation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.
[0353] In a cycle for pair **16** of pads **13** and **14**, there are successively following phases: active phase **101**, unclamping phase **102**, U-turn phase **103**, clamping phase **104**, and again active phase **101** of next cycle.
[0354] In a cycle for pair **15** of pads **11** and **12**, there are successively following phases: active phase **111**, unclamping phase **112**, U-turn phase **113**, clamping phase **114**, and again active phase **111** of next cycle.
[0355] With respect to translation motion of flexible elongated element **10**, this cycle of a pair **16** of pads **13** and **14**, may be seen as including the following phases: active phase **105** translating the flexible elongated medical element **10** with both pads **13** and **14** clamped around this flexible elongated medical element **10**, then unclamping phase **106** with both pads **13** and **14** releasing this flexible elongated medical element **10**, then U-turn phase **107** with pads **11** and **12** getting back towards their initial position, then clamping phase **108** with pads **13** and **14** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **105** of next cycle.
[0356] With respect to translation motion of flexible elongated element **10**, this cycle of a pair **15** of pads **11** and **12**, may be seen as including the following phases: active phase **115** translating the flexible elongated medical element **10** with both pads **11** and **12** clamped around this flexible elongated medical element **10**, then unclamping phase **116** with both pads **11** and **12** releasing this flexible elongated medical element **10**, then U-turn phase **117** with pads **11** and **12** getting back towards their initial position, then clamping phase **118** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **115** of next cycle.
[0357] Considering the FIG. **5**, it can be seen that a hand **15** can move the flexible elongated medical element **10** only during the active phase **111**, i.e. when the two pads **11** and **12** are clamped. In order to obtain a continuous motion of the flexible elongated medical element **10**, both pairs **15** and **16** of pads need to cooperate. When one pair **15** is in U-turn phase **113**, the other pair **16** should be in active phase **101** thus ensuring that the flexible elongated medical element **10** is being clamped and moved by at least one pair of pads at any time.
[0358] The FIG. **11**text missing or illegible when filed illustrates such a cooperation for translation. It shows the translation of both hands **15** and **16** (RT and LT) combined with clamping/unclamping of both hands **15** and **16**. In order to get the best possible cooperation, the curves of one hand should be close to exact phase opposition.
[0359] As shown on the colored bar at the bottom of FIG. **11**, there are periods where one hand **15** or **16** is clamped, respectively periods **109** or **119**, and periods **110** where both hands **15** and **16** are clamped. These latter periods **110**, which are also called “overlapping periods” **110**, are very useful to ensure that the motion of flexible elongated medical element is quite or even perfectly fluid.
[0360] FIG. **12** shows schematically graphs of evolution, as a function of time, of clamping states of the two pairs of movables pads, for a slow rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.
[0361] For pair **16** of pads **13** and **14**, there is a succession of cycles each including succession of

an active phase **121** followed by a U-turn phase **122**.
[0362] For pair **15** of pads **11** and **12**, there is a succession of cycles each including succession of an active phase **123** followed by a U-turn phase **124**.
[0363] With respect to rotation motion of flexible elongated element **10**, this cycle of a pair **16** of pads **13** and **14**, may be seen as including the following phases: active phase **105** rotating the flexible elongated medical element **10** with both pads **13** and **14** clamped around this flexible elongated medical element **10**, then unclamping phase **106** with both pads **13** and **14** releasing this flexible elongated medical element **10**, then U-turn phase **107** with pads **13** and **14** getting back towards their initial position, then clamping phase **108** with pads **13** and **14** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **105** of next cycle.
[0364] With respect to rotation motion of flexible elongated element **10**, this cycle of a pair **15** of pads **11** and **12**, may be seen as including the following phases: active phase **115** rotating the flexible elongated medical element **10** with both pads **11** and **12** clamped around this flexible elongated medical element **10**, then unclamping phase **116** with both pads **11** and **12** releasing this flexible elongated medical element **10**, then U-turn phase **117** with pads **11** and **12** getting back towards their initial position, then clamping phase **118** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **115** of next cycle.
[0365] As seen previously, the same pair of pads, either pair **15** of pads **11** and **12** or pair **16** of pads **13** and **14**, handles simultaneously the translation and rotation of the flexible elongated medical element, thanks to motions along the x and z axis.
[0366] In addition, it can be seen that the translation and rotation speeds are user defined and can be set in a fully independent manner from each other.
[0367] When considering both pads of both hands **15** and **16** during a simultaneous and continuous translation and rotation of a flexible elongated medical element **10**, the consequence of the motion principles of the pads **11** and **12**, or **13** and **14**, along the various axes imply the following rules: [0368] unclamping has to be done when the pads **11** and **12**, or **13** and **14**, reach their maximum position along the x (for translation) or z (for rotation) axis, [0369] unclamping of both hands **15** and **16** has to be avoided since this would stop the flexible elongated medical element **10** motion. Moreover, unclamping of both hands **15** and **16** could lead to unwanted and thus unsafe movements of the flexible elongated medical element **10**.
[0370] The problem lies in the potential contraction between these points. This can be seen when considering both a slow rotation as on FIG. **12**, and a fast translation as on FIG. **13**, with cycles of successive active phase **131** and U-turn phase **132**, for pair **16** of pads **13** and **14**, and cycles of successive active phase **133** and U-turn phase **134**, for pair **15** of pads **11** and **12**.
[0371] FIG. **13** shows schematically graphs of evolution, as a function of time, of clamping states of the two pairs of movables pads, for a fast translation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.
[0372] With respect to translation motion of flexible elongated element **10**, this cycle of a pair **16** of pads **13** and **14**, may be seen as including the following phases: active phase **105** translating the flexible elongated medical element **10** with both pads **13** and **14** clamped around this flexible elongated medical element **10**, then unclamping phase **106** with both pads **13** and **14** releasing this flexible elongated medical element **10**, then U-turn phase **107** with pads **13** and **14** getting back towards their initial position, then clamping phase **108** with pads **13** and **14** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **105** of next cycle.
[0373] With respect to translation motion of flexible elongated element **10**, this cycle of a pair **15** of pads **11** and **12**, may be seen as including the following phases: active phase **115** translating the flexible elongated medical element **10** with both pads **11** and **12** clamped around this flexible

elongated medical element **10**, then unclamping phase **116** with both pads **11** and **12** releasing this flexible elongated medical element **10**, then U-turn phase **117** with pads **11** and **12** getting back towards their initial position, then clamping phase **118** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **115** of next cycle.

[0374] Now, what happens if these two movements are to be combined? The movements along the x and z axis are independent and do not interfere with each other. However, the movements along the y axis (clamping) have a different status. Indeed, the clamping movement is linked to both translation and rotation, because, when unclamping a hand **15**, both the translation and the rotation of this hand **15** are interrupted. Therefore, when translation (for example) has reached its maximum position and requests an unclamping motion, this has a side effect on rotation, which may not need unclamping at this time.

[0375] FIG. **14** shows schematically graphs of evolution, as a function of time, of clamping state of one of the pairs of movables pads, for a combined translation and rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0376] With respect to translation motion of flexible elongated element **10**, this cycle of a pair **16** of pads **13** and **14**, may be seen as including the following phases: active phase **105** translating the flexible elongated medical element **10** with both pads **13** and **14** clamped around this flexible elongated medical element **10**, then unclamping phase **106** with both pads **13** and **14** releasing this flexible elongated medical element **10**, then U-turn phase **107** with pads **13** and **14** getting back towards their initial position, then clamping phase **108** with pads **13** and **14** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **105** of next cycle.

[0377] With respect to rotation motion of flexible elongated element **10**, this cycle of a pair **16** of pads **13** and **14**, may be seen as including the following phases: active phase **145** rotating the flexible elongated medical element **10** with both pads **13** and **14** clamped around this flexible elongated medical element **10**, then unclamping phase **146** with both pads **13** and **14** releasing this flexible elongated medical element **10**, then U-turn phase **147** with pads **13** and **14** getting back towards their initial position, then clamping phase **148** with pads **13** and **14** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **145** of next cycle.

[0378] However, in a first approach, in order to allow both the translation and the rotation to do their respective U-turn phases **107** and **147**, unclamping will be performed if either the translation or the rotation needs it.

[0379] To do so, in FIG. **14**, one clamping curve (RC) in case of rotation only is combined with the same curve in case of translation only. For a combined rotation and translation, these two curves will be combined. Since, in this case, the unclamped position is the lowest, the combination will consist in calculating the minimum value of these two curves.

[0380] The result of this combination is the lowest curve of FIG. **14**, with clamped position **141** and unclamped position **143**, with clamping **144** and unclamping **142**.

[0381] If both rotation and translation ask for unclamp as in zone **140**, then unclamp prevails. If both rotation and translation ask for clamp as in zone **149**, then clamp prevails.

[0382] FIG. **15** shows schematically graphs of evolution, as a function of time, of clamping state of the other one of the pairs of movables pads, for a combined translation and rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0383] With respect to translation motion of flexible elongated element **10**, this cycle of a pair **15** of pads **11** and **12**, may be seen as including the following phases: active phase **115** translating the flexible elongated medical element **10** with both pads **11** and **12** clamped around this flexible elongated medical element **10**, then unclamping phase **116** with both pads **11** and **12** releasing this flexible elongated medical element **10**, then U-turn phase **117** with pads **13** and **14** getting back towards their initial position, then clamping phase **118** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **105** of next cycle.

[0384] With respect to rotation motion of flexible elongated element **10**, this cycle of a pair **15** of pads **11** and **12**, may be seen as including the following phases: active phase **155** rotating the flexible elongated medical element **10** with both pads **11** and **12** clamped around this flexible elongated medical element **10**, then unclamping phase **156** with both pads **11** and **12** releasing this flexible elongated medical element **10**, then U-turn phase **157** with pads **11** and **12** getting back towards their initial position, then clamping phase **158** with pads **11** and **12** getting closer until touching and maintaining this flexible elongated medical element **10**, and then again active phase **155** of next cycle.

[0385] The result of this combination is the lowest curve of FIG. **14**, with clamped position **151** and unclamped position **153**, with clamping **154** and unclamping **152**.

[0386] If the RC curve of clamping in case of combined translation and rotation from FIG. **14**, and the LC curve of clamping in case combined translation and rotation from FIG. **15** are now extracted, this leads to get at FIG. **16**.

[0387] FIG. **16** shows schematically graphs of evolution, as a function of time, of the wanted clamping state of both pairs of movables pads, for a combined translation and rotation of the flexible elongated medical device, in an example of a catheter robot module according to the invention.

[0388] In this FIG. **16**, the periods of time **160** when both the left hand **15** and the right hand **16** are unclamped (RC and LC low) have been highlighted. This is a problem since this corresponds to both pairs **15** and **16** simultaneously unclamped, what is to be avoided, since these periods **160** lead to loss of control over flexible elongated medical element **10** and to idle periods **160** where no motion can be imparted to flexible elongated medical element **10**, whether translation motion or rotation motion.

[0389] During combined translation and rotation, the path of the pads along the x and z axis cannot behave independently without inconvenience, only considering the user speed setpoint. But, on the contrary, the translation (along the x axis) has to take into account what is happening on rotation (along the z axis), and vice-versa.

[0390] FIG. **17** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a sudden change of user speed setpoint, in an example of a catheter robot module according to the invention.

[0391] The translation of the pair of pads of one hand should be in phase opposition with the pads of the other hand (RT and LT). The same applies for rotation (RR and LR).

[0392] This should remain true even in the case of user speed setpoint change.

[0393] An apparently simple solution to this problem could be the following: the speed during the U-turn phase should change proportionally to the speed of the active phase, this latter speed being determined by the user). As can be seen on FIG. **17**, the coordinated motion of the pads of the two hands **16** and **15** for translation (RT and LT).

[0394] For pair **16** of pads **13** and **14**, active phase **171** is followed by U-turn phase **172**.

[0395] For pair **15** of pads **11** and **12**, active phase **173** is followed by U-turn phase **174**.

[0396] In the middle of the curve, at time **170**, the user changes the speed setpoint from a slower value **178** to a faster value **179**. There is a corresponding change in slope **175** from smoother slope **176** to steeper slope **177**, and the period of the cycles becomes shorter.

[0397] Using a U-turn speed which is proportional to the active phase speed results in a synchronization between hands **16** and **15** that is maintained after the user speed setpoint change.

[0398] However, this method would have two problems.

[0399] The first problem would be related to the lower speeds. As already seen earlier, when the speed in the U-turn phase **172** or **174** is proportional to the speed in the active phase **171** or **173**, with for example a slow rotation speed which is combined with a fast translation speed, a clamping conflict is created, and thus, in short, it doesn't work efficiently.

[0400] The second problem would be related to higher speeds. In this case, the actuators will be limited, because very high actuator speeds and acceleration imply bigger actuators with more heat dissipation. In order to keep dimensions and cooling measures within reasonable limits, it is needed to put a maximum speed limit for the U-turn phase **172** or **174**.

[0401] Those two problems can be merged into one statement: it is hard and it may even become not possible to set the U-turn speed as a value being proportional to the active speed, at least not without great inconvenience.

[0402] FIG. **18** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a change of user speed setpoint, with a desynchronization problem with a fixed duration of U-turn, in an example of a catheter robot module according to the invention.

[0403] A tentative solution could be to use a constant U-turn speed. This could be set it as high as possible, in a tentative to minimize the clamping conflict. The curves of the FIG. **17** would then change into those of FIG. **18**.

[0404] However, this results in a synchronization loss between the hands: at the beginning, the hands **16** and **15** are synchronized (t.sub.1=t.sub.2), but after the user speed setpoint change, synchronization between both hands **16** and **15** is lost (t′.sub.1 ≠=t′.sub.2).

[0405] Therefore, there is a need for a mechanism to control and maintain synchronization between the two hands **16** and **15**.

[0406] FIG. **19** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with a first margin, in an example of a catheter robot module according to the invention.

[0407] In the cycle, active phase **191** is followed by U-turn phase **192**. Extension of the travel range along x axis, direction of translation of flexible elongated medical element **10**, is usually a standard range **195**, but in some cases it can travel within a maximum range **196** which adds a margin, split into two half margins, upper half margin **193** and lower half margin **194**, added at each end of the standard range **195**.

[0408] There is a similar margin for extension of the travel range along z axis, axis used to impart rotation to the flexible elongated medical element **10**.

[0409] In order to avoid any clamping conflict, the travel range along the x and z axis is split between a standard range **195** and a maximum range **196**, the difference between the two being the margin. Consider the curve for one movement, for example RT. In this example, the standard range **195** is used:

[0410] FIG. **20** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with use of the first margin, with a positive user setpoint, in an example of a catheter robot module according to the invention.

[0411] This gives a margin for the translation. When reaching the maximum value of the standard range **195**, if the other hand is clamped, then the U-turn can be started normally. But if the other hand is unclamped, then the upper half margin **193** between the maximum range **196** and the standard range **195** can be used in order to give time to the other hand to finish its U-turn phase, as can be seen on FIG. **20**.

[0412] In this FIG. **20**, can be seen such a “special” cycle where the pads travel further along the x axis in order to give time to the other hand doing its U-turn. Then the U-turn is done, with two key characteristics:

[0413] The travel range of U-turn phase is always the same, which means that is has the same amplitude as the amplitude of the standard range **195**,

[0414] The duration of U-turn phase is defined to keep synchronization between hands **16** and **15**, as is explained below.

[0415] As a result, the hands synchronization is maintained here, because the RT signal returns to its original path once the "special" cycle is finished.
[0416] Therefore, this enables to solve the clamping conflict without interfering with the hand synchronization control, which is now explained below.
[0417] The margin has to be dimensioned so as to avoid a situation where the clamping conflict would still not be solved when the pad reaches the maximum position of the maximum range **196**. This will depend on system parameters: maximum translation speed, maximum rotation speed, minimum and maximum U-turn speed (for translation and rotation), etc. . . .
[0418] In FIG. **8**, it can be seen that unclamping has to start before the pad reaches its maximum position. This small delay is needed by the response time of the clamping/unclamping actuators. Therefore, the decision to start a U-turn has also to be anticipated by the same delay.
[0419] FIG. **21** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with use of the first margin, with a negative user setpoint, in an example of a catheter robot module according to the invention.
[0420] The FIG. **20** illustrates the case of a positive user speed setpoint. The margin is split between a lower half margin **194** and an upper half margin **193**. In this case, only the upper half margin **193** is used. In case of a negative used speed setpoint, the lower half margin **194** is used, as can be seen on FIG. **21**.
[0421] FIG. **22** shows schematically a graph of evolution, as a function of time, of translation of the flexible elongated medical device, with use of a second margin, with a positive user setpoint, in an example of a catheter robot module according to the invention.
[0422] To summarize, the system has a tolerance on U-turn phase starting time, allowing to have a delay on it, compared to the ideal starting time.
[0423] In another embodiment, we can also allow the system to start the U-turn phase in advance. When considering the movements of one hand, the movements along the x (translation) and z (rotation) axis are usually not synchronized, as can be seen for example on FIGS. **12** and **13**, therefore leading to potentially longer unclamped periods, as can be seen on FIG. **15**, which, in turn, could cause a clamping conflict, as shown on FIG. **16**. Being able to anticipate a U-turn would then help limiting the unclamped periods. If, for example, the z movement starts its U-turn, and therefore triggers an unclamping of the hand, and, at the same time, the x movement is clamped but close "enough" to the end of its range, i.e. the point where it would also start a U-turn, it could then be wiser for the x movement to take advantage of the fact that the hand is unclamped to start its own U-turn, thus reducing the overall time during which the hand is unclamped.
[0424] This is shown on FIG. **22**, where the notion of minimum range **229** has been added. Therefore, between the minimum range **229** and the standard range **225**, the rule could become to start a U-turn if the hand is already unclamped (due to the other movement, i.e. x for z, or z for x). The behavior between the standard range **225** and the maximum range **226** remains the same as described previously. In the FIG. **22**, the minimum range **229** is equal to the standard range **225** minus a first margin (being the sum of upper half margin **223** and lower half margin **224**), and the maximum range **226** is equal to the standard range **225** plus a second margin (being the sum of upper half margin **227** and lower half margin **228**). First margin and second margin are equal on FIG. **22**, but they could also be different from each other.
[0425] FIG. **23** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with synchronized and unsynchronized moves, in an example of a catheter robot module according to the invention.
[0426] Pair **16** of pads **13** and **14** follows a cycle of active phase **231** and U-turn phase **232**.
[0427] Pair **15** of pads **11** and **12** follows a cycle of active phase **233** and U-turn phase **234**.
[0428] Either pairs **16** and **15** are synchronized with each other and their paths crossings **235** are periodical and happen at half travel extension, or pairs **16** and **15** are not synchronized with each other and their paths crossings **236** are not periodical and do not happen at half travel extension.

[0429] Adapting the synchronization between hands **16** and **15** could be compared with a phase lock loop control (PLL) mechanism, aiming at controlling the phase of a slave signal from the phase of master signal. In short, the algorithm for the slave signal could more or less amount to the following: "if you are late, accelerate, if you have advance, slow down".
[0430] The speed of the pads along the x (translation) and z (rotation) axis during the active phases **231** or **233** cannot be chosen because there are imposed by the user speed settings. However, the U-turn speed can be chosen. The travel range of the U-turn phase has to be kept constant, as seen above, so the U-turn duration will be variated to control synchronization between pairs **16** and **15**.
[0431] Therefore, when needed, "accelerate" or "slow down" will be based on adjustments of the U-turn duration.
[0432] Each time a hand **16** or **15** starts a U-turn phase, it acts as a slave that has to synchronize itself with the other hand **15** or **16**, which is then the master. So, the master-slave scheme is different here, because each hand **16** or **15** acts alternatively as a master and a slave.
[0433] Calculation of this U-turn duration will now be described below.
[0434] First of all, when observing synchronized (paths crossings **235**) vs unsynchronized signal (paths crossings **236**), it can be seen that in the first synchronized case the curves cross always at the same value (paths crossings **235**), whereas unsynchronized curves cross alternatively at a higher and lower level (paths crossings **236**).
[0435] FIG. **24** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a first step of correction of desynchronization, in an example of a catheter robot module according to the invention.
[0436] When starting from an unsynchronized situation (paths crossings **236**), as illustrated on FIG. **24**, it can be seen that LT curve has reached its maximum value, and that there is no clamping conflict so that the U-turn phase can now start. t.sub.U-turn has to be now calculated. Different values would yield different paths, as represented in different dotted lines **237** or **238** or **239**.
[0437] FIG. **25** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a second step of correction of desynchronization, in an example of a catheter robot module according to the invention.
[0438] First of all, an extrapolation can be done as to what the RT curve should look like during the next cycle: dotted line **251** for the rest of active phase and then dotted line **252** for the U-turn phase with a supposed path crossing **255** at middle travel extension with the LT curve.
[0439] FIG. **26** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a third step of correction of desynchronization, in an example of a catheter robot module according to the invention.
[0440] For that, the t.sub.U-turn value of the next cycle is also extrapolated, which is called t.sub.ideal U-turn. Calculation of t.sub.ideal U-turn will be explained in more detail below.
[0441] The next active phase of the LT curve can now be drawn by dotted line **253**, because the slope is known, since it only depends on the user speed setpoint, and it has to cross the supposed path crossing **255** because this would allow for resynchronization between both pairs **16** and **15**, via the resynchronization of LT curve with the RT curve.
[0442] FIG. **27** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a fourth step of correction of desynchronization, in an example of a catheter robot module according to the invention.
[0443] Linking existing segments **233** and **253** will give the path of the U-turn phase **254** and therefrom the value of t.sub.U-turn can be got directly (projection on time axis to the travel extension of U-turn phase **254**).
[0444] FIG. **28** shows schematically a graph of evolution, as a function of user speed setpoint, of a wished U-turn duration, in an example of a catheter robot module according to the invention.
[0445] To calculate the next t.sub.ideal U-turn, an arbitrary function depending only on the user translation and rotation speed setpoints is used. In other words, it is the t.sub.U-turn value that

would be got in case of no need to re-synchronize the two hands **16** and **15**. This function can be chosen in an arbitrary manner. An example of embodiment could be the one as represented on FIG. **28**.

[0446] In this example, the faster the user speed setpoint, the shorter t.sub.ideal U-turn. Indeed, for a high speed, it is needed to do the U-turn fast.

[0447] In the curved part **280**, t.sub.ideal U-turn is inversely proportional to the user speed setpoint. This means that the U-turn speed will be proportional to the user speed setpoint.

[0448] The two horizontal parts **281** and **282** have specific respective functions. In the lower user speed setpoint values, lower horizontal part **282** prevents the algorithm from using too high t.sub.ideal U-turn values, which would lead to long periods with one hand unclamped and thus difficulties to handle clamping conflicts. On the other side of the curve, thanks to the upper horizontal part **281**, there is a minimum t.sub.ideal U-turn value that is needed because the actuators are limited in speed and could not follow, at least not easily, the requested path if t.sub.ideal U-turn were too small.

[0449] Other curves are possible but they have to be decaying as a function of the user speed setpoint.

[0450] FIG. **29** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a step of management of clamping conflict, in an example of a catheter robot module according to the invention.

[0451] In case the last RT cycle had a clamping conflict and thus used more amplitude than the standard range, the position of the path crossing **236** using an extrapolated RT curve (dotted line **292**) has to be calculated, i.e. what the RT curve would have been in case of no conflict.

[0452] FIG. **30** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with another step of management of clamping conflict, in an example of a catheter robot module according to the invention.

[0453] A similar extrapolation is also needed in case the current LT cycle has a conflict. So, in case the last LT cycle had a clamping conflict and thus used more amplitude than the standard range, the position of the path crossing using an extrapolated LT curve (dotted line **302**) has to be calculated, i.e. what the LT curve would have been in case of no conflict.

[0454] Now that t.sub.U-turn has been calculated, a minimum value and a maximum value have to be applied. Indeed, the calculation steps detailed above do not prevent from excessively small or high values, which may have same drawbacks as the ones explained above to justify the horizontal parts **281** and **282** on FIG. **28**. In a preferred embodiment, the minimum value is chosen smaller than the minimum of the function represented on FIG. **28**, and the maximum greater than its function maximum.

[0455] If those minimum and maximum limits have been hit, it can be seen that the resulting t.sub.U-turn value will then not provide synchronization immediately back, because in this case several cycles will be needed to catch up.

[0456] The term “user speed setpoint” has been used without specifying whether it is translation or rotation. Actually, it combines both. In case of a combined translation and rotation, the t.sub.U-turn value is calculated using the function of FIG. **28** (or any other embodiment of such as function), for both translation and rotation, yielding two t.sub.U-turn values: t.sub.U-turn_T and t.sub.U-turn_R.

[0457] The minimum of the two: t.sub.U-turn=min(t.sub.U-turn_T, t.sub.U-turn_R) is then used.

[0458] This is because the same unclamping movement is used for translation and rotation. A slow active phase can be combined with a fast U-turn phase, but the opposite is false, because it would cause relatively long unclamping period and create more clamping conflict possibilities. In an extreme situation where the U-turn phase would be longer than the active phase, this could run the risk to lead to having a permanent clamping conflict.

[0459] The whole synchronization process could be summarized roughly as follows. The left hand **15** synchronizes on right hand **16**, then the right hand **16** synchronizes on left hand **15**, etc . . . This process is repeated indefinitely and could lead to some instabilities, the two hands **16** and **15** "fighting" against each other. As a result, synchronization would run the risk of never being fully obtained and could somewhat oscillate between "advance" and "delay". In order to overcome this problem, in a preferred embodiment, only a fraction of the correction could be applied. t.sub.calculated U-turn the t.sub.U-turn value are calculated according to the method detailed above. Then, the correction factor is: t.sub.correction=t.sub.calculated U-turn−t.sub.ideal U-turn. Applying part of the correction then means using this formula: the t.sub.U-turn=t.sub.ideal U-turn+α. t.sub.correction, where a is an arbitrary factor: 0<α≤1. Any value of a is possible. Lower values give better stability, higher values require less cycles to get synchronization back. In a preferred embodiment α=½ Is chosen. This calculation is to be applied before the last step where t.sub.U-turn is limited between minimum and maximum values.

[0460] For sake of clarity, it has been assumed on many figures, that the user speed setpoints are constant. In practice, this may not be the case, as the user could frequently change translation and rotation speed of the manipulated flexible elongated medical elements. By the way, Human-Man Interfaces allow the user to change those speed in a continuous way.

[0461] FIG. **31** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a progressive change of user speed setpoint, in an example of a catheter robot module according to the invention.

[0462] For pair **16** of pads **13** and **14**, active phase **311** is followed by U-turn phase **312**. In the middle of the curve, during a period **318**, the user progressively changes the speed setpoint from a slower value **317** to a faster value **319**. There is a corresponding progressive change in slope of active phase **313**, past cycles presenting an active phase **311** and a U-turn phase **312** with a smoother slope, whereas future cycles will present an active phase **315** and a U-turn phase **316** with steeper slope, the period of the cycles becoming shorter.

[0463] FIG. **32** shows schematically graphs of evolution, as a function of time, of translation of the flexible elongated medical device, with a change of user speed setpoint during the U-turn, in an example of a catheter robot module according to the invention.

[0464] For pair **16** of pads **13** and **14**, active phase **231** is followed by U-turn phase **232**.

[0465] For pair **15** of pads **11** and **12**, active phase **233** is followed by U-turn phase **234**.

[0466] In the middle of the curve, at a time **328**, the user suddenly changes the speed setpoint from a slower value **327** to a faster value **329**.

[0467] There is a corresponding sudden change in slope of U-turn phase **234**, past cycles presenting: [0468] for pair **16**, an active phase **231** and a U-turn phase **232** with a steeper slope, whereas future cycles will present an active phase **321** and a U-turn phase **322** with smoother slope, the period of the cycles becoming longer, [0469] for pair **15**, an active phase **233** and a U-turn phase **234** with a steeper slope, whereas future cycles will present an active phase **323** and a U-turn phase **324** with smoother slope, the period of the cycles becoming longer, [0470] both pairs **16** and **15** becoming resynchronized, as can be seen through synchronized new path crossing **325**, whereas before they were desynchronized as could be seen through desynchronized old path crossing **236**.

[0471] On FIG. **31**, a user speed setpoint variation is applied to the curves, if this variation takes place during the active phase **311**, then the slope of the curve has just to be modified accordingly.

[0472] If this change occurs during a U-turn phase **312**, the same calculation steps (as already shown with respect to FIGS. **25**, **26** and **27**) can be used, except that the extrapolated curves will have to take into account the new setpoints. This is illustrated in FIG. **32**, in case of a reduction of the translation speed setpoint.

[0473] The two mechanisms used to solve the double problem of phase opposition control and of synchronization conflict have been explained above.

[0474] Those two mechanisms are applied to each pair of pads, either pair **16** of pads **13** and **14**, or pair **15** of pads **11** and **12**, and to each type of motion, i.e. generating the translation and the rotation of the flexible elongated medical element **10**.

[0475] This means that those two mechanisms are applied simultaneously to **4** motions indeed:
[0476] The motion of the first pair **15** of pads **11** and **12** for the translation of the flexible elongated medical element **10**, [0477] The motion of the first pair **15** of pads **11** and **12** for the rotation of the flexible elongated medical element **10**, [0478] The motion of the second pair **16** of pads **13** and **14** for the translation of the flexible elongated medical element **10**, [0479] The motion of the second pair **16** of pads **13** and **14** for the rotation of the flexible elongated medical element **10**.
[0480] This is obtained thanks to **4** Finite States Machines (FSM). Each FSM has to: [0481] Ensure that, when clamped (i.e. during the active phase), the pads have a linear motion and impart the correct speed to the flexible elongated medical element **10** (i.e. the user setpoint speed), [0482] Choose when to quit the active phase and start the U-Turn phase, using the "margin mechanism" as explained in FIGS. **19** to **22**. [0483] Determine the initial U-turn duration, and thus the U-turn speed, using the "U-turn duration adaptation mechanism" as explained above. [0484] Adapt the U-turn speed in real-time during the U-turn phase, by refreshing the U-turn speed calculation, using the same "U-turn duration adaptation mechanism" as explained above.
[0485] The table 2 below lists the characteristics of these 4 FSM:
TABLE-US-00002 TABLE 2 FSM name Pair of pads Motion of flexible FSM.sub.RT FSM.sub.RR FSM.sub.LT FSM.sub.LR elongated medical Right Right Left Left element Translation Rotation Translation Rotation State variable State.sub.RT State.sub.RR State.sub.LT State.sub.LR State duration T.sub.RT T.sub.RR T.sub.LT T.sub.LR Position x.sub.R z.sub.R x.sub.L z.sub.L Clamping y.sub.R y.sub.L
[0486] The state of each FSM seems to be defined by what would appear to be "3½" variables. In reality, these variables are indeed: [0487] The state; until here, we have described two states "active" and "U-turn" in a first approximation. As explained below, there are indeed more states within a whole cycle, [0488] The duration since the current state is entered; depending of the embodiment, this variable may be or may not be used. [0489] The position of one pad along the x or z axis, depending on which motion the FSM controls, [0490] The position of one pad along the y axis, which controls the clamping status (clamped/unclamped). This is the "half variable" because it is shared between two FSM. For example, the clamping status of the left hand (y.sub.L) **15** is related to both FSM.sub.LT and FSM.sub.LR. This shared variable is a consequence of the fact that unclamping a pair of pads unclamps both the motion along the x and z axis, i.e. stops the translation and the rotation of the flexible elongated medical element **10**. In case of disagreement between two FSM regarding y, the priority is given to the unclamping command over the clamping command.
[0491] Each FSM changes every Δt. Δt is chosen according to the wanted reactivity of the system and to the actuators characteristics. In a preferred embodiment, Δt=1 ms.
[0492] V.sub.T is called the user translation speed setpoint and V.sub.R the user rotation speed setpoint (In V.sub.R, "R" means "rotation". In X.sub.R, Y.sub.R and Z.sub.R, it means "right").
[0493] FIG. **33** shows schematically the temporal evolution of four finite state machines, in an example of a catheter robot module according to the invention.
[0494] The temporal evolution of the four FSM **331**, **332**, **333** and **334**, is described in the FIG. **33**.
[0495] In finite state machine (FSM) **331**, dedicated to right translation, i.e. translation of pair **16** of pads **13** and **14**: [0496] Inputs are (top-down) at time t: [0497] User translation speed setpoint V.sub.T(t), [0498] User rotation speed setpoint V.sub.R(t), [0499] Type of state, for translation of pair **16** of pads **13** and **14**, State.sub.RT(t), [0500] Duration of said state, for translation of pair **16** of pads **13** and **14**, T.sub.RT(t), [0501] x position, for pair **16** of pads **13** and **14**, X.sub.R(t), [0502] clamping, for pair **16** of pads **13** and **14**, Y.sub.R(t), [0503] clamping, for pair **15** of pads **11** and **12**, Y.sub.L(t), [0504] Outputs are (top-down) at time t+Δt: [0505] Type of state, for translation of pair **16** of pads **13** and **14**, State.sub.RT(t+Δt), [0506] Duration of said state, for right translation, T.sub.RT(t+Δt), [0507] x position, for pair **16** of pads **13** and **14**, X.sub.R(t+Δt), [0508] clamping, for pair **16** of pads **13** and **14**, Y.sub.R(t+Δt).

[0509] In finite state machine (FSM) **332**, dedicated to right rotation, i.e. rotation of pair **16** of pads **13** and **14**: [0510] Inputs are (top-down) at time t: [0511] User translation speed setpoint V.sub.T(t), [0512] User rotation speed setpoint V.sub.R(t), [0513] Type of state, for rotation of pair **16** of pads **13** and **14**, State.sub.RR(t), [0514] Duration of said state, for rotation of pair **16** of pads **13** and **14**, T.sub.RR(t), [0515] z position, for pair **16** of pads **13** and **14**, Z.sub.R(t), [0516] clamping, for pair **16** of pads **13** and **14**, Y.sub.R(t), [0517] clamping, for pair **15** of pads **11** and **12**, Y.sub.L(t), [0518] Outputs are (top-down) at time t+Δt: [0519] type of state, for rotation of pair **16** of pads **13** and **14**, State.sub.RR(t+Δt), [0520] Duration of said state, for rotation of pair **16** of pads **13** and **14**, T.sub.RR(t+Δt), [0521] z position, for pair **16** of pads **13** and **14**, Z.sub.R(t+Δt), [0522] clamping, for pair **16** of pads **13** and **14**, Y.sub.R(t+Δt).

[0523] In finite state machine (FSM) **333**, dedicated to left translation, i.e. translation of pair **15** of pads **11** and **12**: [0524] Inputs are (top-down) at time t: [0525] User translation speed setpoint V.sub.T(t), [0526] User rotation speed setpoint V.sub.R(t), [0527] Type of state, for translation of pair **15** of pads **11** and **12**, State.sub.LT(t), [0528] Duration of said state, for translation of pair **15** of pads **11** and **12**, T.sub.LT(t), [0529] x position, for pair **15** of pads **11** and **12**, X.sub.L(t), [0530] clamping, for pair **15** of pads **11** and **12**, Y.sub.L(t), [0531] clamping, for pair **16** of pads **13** and **14**, Y.sub.R(t), [0532] Outputs are (top-down) at time t+Δt: [0533] Type of state, for translation of pair **15** of pads **11** and **12**, State.sub.LT(t+Δt), [0534] Duration of said state, for right translation, T.sub.LT(t+Δt), [0535] x position, for pair **15** of pads **11** and **12**, X.sub.L(t+Δt), [0536] clamping, for pair **15** of pads **11** and **12**, Y.sub.L(t+Δt).

[0537] In finite state machine (FSM) **334**, dedicated to left rotation, i.e. rotation of pair **15** of pads **11** and **12**: [0538] Inputs are (top-down) at time t: [0539] User translation speed setpoint V.sub.T(t), [0540] User rotation speed setpoint V.sub.R(t), [0541] Type of state, for rotation of pair **15** of pads **11** and **12**, State.sub.LR(t), [0542] Duration of said state, for rotation of pair **15** of pads **11** and **12**, T.sub.LR(t), [0543] z position, for pair **15** of pads **11** and **12**, Z.sub.L(t), [0544] clamping, for pair **15** of pads **11** and **12**, Y.sub.L(t), [0545] clamping, for pair **16** of pads **13** and **14**, Y.sub.R(t), [0546] Outputs are (top-down) at time t+Δt: [0547] type of state, for rotation of pair **15** of pads **11** and **12**, State.sub.LR(t+Δt), [0548] Duration of said state, for rotation of pair **15** of pads **11** and **12**, T.sub.LR(t+Δt), [0549] z position, for pair **15** of pads **11** and **12**, Z.sub.L(t+Δt), [0550] clamping, for pair **15** of pads **11** and **12**, Y.sub.L(t+Δt).

[0551] FIG. **34** shows schematically the **12** states of a cycle of the four finite state machines, in an example of a catheter robot module according to the invention.

[0552] All the states for FSM.sub.RT **331** (i.e. the possible values of State.sub.RT) are now described. Man skilled in the art will easily transpose for the 3 other FSM **332**, **333** and **334**.

[0553] There is a total of 12 states. These 12 states are split into two groups: the states 1 to 6 correspond to a positive user setpoint translation speed and the states 7 to 12 to a negative user setpoint translation speed: [0554] state **341** UP_LINEAR: this is the active phase. [0555] state **342** UP_FROZEN: the motion of the pads, and thus the flexible elongated medical element, is stopped. [0556] state **343** UP_UNCLAMP: continue to impart a linear motion along the x axis according to the user speed setpoint, while sending an “unclamp” command the pads along the y axis. [0557] state **344** UP_DOWN: this the U-turn phase. [0558] state **345** UP_WAIT_CLAMP: during this state, the speed of the pair of pads along the x axis should be equal to the user translation speed setpoint, as in the 1-UP_LINEAR phase. [0559] state **346** UP_CLAMP: during this state, the pads are re-clamped while maintaining the speed of the pads along the x axis equal to the user translation speed setpoint. [0560] state **347** DOWN_LINEAR: this is the active phase. [0561] state **348** DOWN _FROZEN: the motion of the pads, and thus the flexible elongated medical element, is stopped. [0562] state **349** DOWN_UNCLAMP: continue to impart a linear motion along the x axis according to the user speed setpoint, while sending an “unclamp” command the pads along the y axis. [0563] state **350** DOWN_UP: this the U-turn phase. [0564] state **351** DOWN_WAIT_CLAMP: during this state, the target speed of the pair of pads along the x axis should be equal to the user translation speed setpoint, as in the 7-DOWN_LINEAR phase. [0565] state **352** DOWN _CLAMP: during this state, the pads are re-clamped while maintaining the speed of the pads along the x axis equal to the user translation speed setpoint.

[0566] FIG. **35** shows schematically a synoptic representing a cycle of the four finite state machines **331**, **332**, **333** and **334**, in an example of a catheter robot module according to embodiments of the invention.

[0567] The state **347** is the equivalent of state **341** in case of a negative value, etc. . . . Therefore, there is a detailed explanation for the states **341** to **346** which is similar to the one given for states **347** to **352**.

[0568] States **341** to **352** are: [0569] state **341** UP_LINEAR: this is the active phase. The target speed of the pair of pads along the x axis is equal to the user translation speed setpoint. The margin mechanism is active. This mechanism uses, as input data, the final position of the next phase **343** (UP_UNCLAMP), if switching to this phase now. Let's call this position.sub.final **403**. If position.sub.final **403** has reached the maximum value of the standard range, then switch to the UP_UNCLAMP state **343**, if the other (left) hand is clamped. If the other hand is not clamped, remain in the UP_LINEAR state **341** until position.sub.final reaches the maximum value of the maximum range. If the other hand is still unclamped at this point **400**, then switch to UP_FROZEN state **342**. This case is an emergency measure in case the margin has not been enough to solve a clamping conflict: a well-designed system should never enter in this state and go directly to UP_UNCLAMP state **343** instead.

[0570] The pads should normally be clamped during the UP_LINEAR phase **341**. However, since the clamping status is shared between translation and rotation, FSM.sub.RR may have decided to unclamp **400** the pads. During this unclamping time, the motion of the pads along the x axis (for translation) will continue normally, although it will have of course no effect on the flexible elongated medical element: the translation will then be obtained thanks to other pair of pads.

[0571] If the user translation speed setting becomes negative **401** during this phase, then switch to DOWN_LINEAR **347**. [0572] state **342** UP_FROZEN: the motion of the pads, and thus the flexible elongated medical element, is stopped. As soon as the other hand is clamped again **400**, switch to UP_UNCLAMP **343**.

[0573] In this degraded mode, the wanted motion cannot be imparted to the flexible elongated medical element. However, we still clamp it, in order to ensure, at least, that the flexible elongated medical element handled by the system and cannot move freely in the patient, which would be dangerous.

[0574] If the user translation speed setting becomes negative **401** during this phase, then switch to phase DOWN_LINEAR **347**. There is no switching to DOWN_FROZEN **348** because, in this case, the situation is different: the position is close to the maximum and the position is to be diminished. Therefore, there is no need to unclamp and no more conflict. [0575] state **343** UP_UNCLAMP: continue to impart a linear motion along the x axis according to the user speed setpoint, while sending an “unclamp” command the pads along the y axis. If the pads were already unclamped due to FSM.sub.RR, then nothing has to be done. In a preferred embodiment, the UP_UNCLAMP duration is fixed. This makes the calculation of position.sub.final during the UP_LINEAR state **341** possible. In another preferred embodiment, the UP_UNCLAMP travel range is fixed. Those two embodiments are interesting but other embodiments can be put in place as long as the calculation is possible during the UP_LINEAR state **341** (earlier in time).

[0576] When the UP_UNCLAMP state **343** is finished **410**, switch to UP_DOWN **344**.

[0577] If the user translation speed setting becomes negative during this phase **401**, then switch to phase DOWN_CLAMP **352**. [0578] state **344** UP_DOWN: this the U-turn phase. The pads have to be unclamped during this phase. The “U-turn duration adaptation mechanism” is active during this phase, in order to determine its duration. This mechanism must be active so as to refresh constantly the duration, which can vary according to the user speed setpoints.

[0579] When the UP_DOWN state **346** is finished **420** (the calculated remaining duration is zero or negative) then switch to UP_WAIT_CLAMP **345**.
[0580] If the user translation speed setting becomes negative during this phase **401**, then switch to DOWN_UP **350**. [0581] state **345** UP_WAIT_CLAMP: during this state, the speed of the pair of pads along the x axis should be equal to the user translation speed setpoint, as in the **341** UP_LINEAR phase. However, due to the actuators and the mechanics, the system has a response time to switch from the UP_DOWN **344** to the UP_LINEAR **341** speed. Therefore, during this phase, the actuators receive control signals to obtain a linear motion at the UP_LINEAR speed, but the pads have not reached that speed yet. At this stage, it is thus too early to re-clamp: the pads have to remain unclamped.
[0582] In a preferred embodiment, the UP_WAIT_CLAMP duration is fixed. In another preferred embodiment, the UP_WAIT_CLAMP travel range is fixed. In a third preferred embodiment, the duration is variable, and the state can be quit when the actual speed of the pads is close enough to the wanted speed. In a preferred embodiment, “enough” could be defined as the absolute value of the difference in percentage being lower than a predetermined threshold.
[0583] When the state is finished **430**, according to one of the criteria explained above, switch to UP_CLAMP **346**.
[0584] If the user translation speed setting becomes negative during this phase **401**, then switch to DOWN_UNCLAMP **349**. [0585] state **346** UP_CLAMP: during this state, the pads are re-clamped while maintaining the speed of the pads along the x axis equal to the user translation speed setpoint.
[0586] In a preferred embodiment, the UP_CLAMP duration is fixed. In another preferred embodiment, the UP_CLAMP travel range is fixed. In a third preferred embodiment, the duration is variable, and the state can be quit when the position of the pads along the y axis has reached a predetermined value. In a fourth preferred embodiment the duration is variable, and the state can be quit when the push force applied to the pads along the y axis has reached a predetermined value.
[0587] When the state is finished **410**, according to one of the criteria explained above, switch to UP_LINEAR **341**.
[0588] If the user translation speed setting becomes negative during this phase **401**, then switch to DOWN_UNCLAMP **349**. [0589] state **347** DOWN_LINEAR: this is the active phase. The target speed of the pair of pads along the x axis is equal to the user translation speed setpoint. The margin mechanism is active. This mechanism uses, as input data, the final position of the next phase **349** (DOWN_UNCLAMP), if switching to this phase now. Let's call this position.sub.final **404**. If position.sub.final **404** has reached the minimum value of the standard range, then switch to the DOWN_UNCLAMP state **349**, if the other (left) hand is clamped. If the other hand is not clamped, remain in the DOWN_LINEAR state **347** until position.sub.final reaches the minimum value of the maximum range. If the other hand is still unclamped at this point **400**, then switch to DOWN_FROZEN state **348**. This case is an emergency measure in case the margin has not been enough to solve a clamping conflict: a well-designed system should never enter in this state and go directly to DOWN_UNCLAMP state **349** instead.
[0590] The pads should normally be clamped during the DOWN_LINEAR phase **347**. However, since the clamping status is shared between translation and rotation, FSM.sub.RR may have decided to unclamp **400** the pads. During this unclamping time, the motion of the pads along the x axis (for translation) will continue normally, although it will have of course no effect on the flexible elongated medical element: the translation will then be obtained thanks to other pair of pads.
[0591] If the user translation speed setting becomes positive **402** during this phase, then switch to UP_LINEAR **341**. [0592] state **348** DOWN_FROZEN: the motion of the pads, and thus the flexible elongated medical element, is stopped. As soon as the other hand is clamped again **400**, switch to DOWN_UNCLAMP **349**.
[0593] In this degraded mode, the wanted motion cannot be imparted to the flexible elongated

medical element. However, we still clamp it, in order to ensure, at least, that the flexible elongated medical element handled by the system and cannot move freely in the patient, which would be dangerous.
[0594] If the user translation speed setting becomes positive **402** during this phase, then switch to phase UP_LINEAR **341**. There is no switching to UP_FROZEN **342** because, in this case, the situation is different: the position is close to the minimum and the position is to be increased. Therefore, there is no need to unclamp and no more conflict. [0595] state **349** DOWN_UNCLAMP: continue to impart a linear motion along the x axis according to the user speed setpoint, while sending an "unclamp" command the pads along the y axis. If the pads were already unclamped due to FSM.sub.RR, then nothing has to be done. In a preferred embodiment, the DOWN_UNCLAMP duration is fixed. This makes the calculation of position.sub.final during the DOWN_LINEAR state **347** possible. In another preferred embodiment, the DOWN_UNCLAMP travel range is fixed. Those two embodiments are interesting but other embodiments can be put in place as long as the calculation is possible during the DOWN_LINEAR state **347** (earlier in time).
[0596] When the DOWN_UNCLAMP state **349** is finished **410**, switch to DOWN _UP **350**.
[0597] If the user translation speed setting becomes positive during this phase **402**, then switch to phase UP_CLAMP **356**. [0598] state **350** DOWN _UP: this the U-turn phase. The pads have to be unclamped during this phase. The "U-turn duration adaptation mechanism" is active during this phase, in order to determine its duration. This mechanism must be active so as to refresh constantly the duration, which can vary according to the user speed setpoints.
[0599] When the DOWN_UP state **350** is finished **420** (the calculated remaining duration is zero or negative) then switch to DOWN_WAIT_CLAMP **351**.
[0600] If the user translation speed setting becomes positive during this phase **402**, then switch to UP_ DOWN **344**. [0601] state **351** DOWN_WAIT_CLAMP: during this state, the target speed of the pair of pads along the x axis should be equal to the user translation speed setpoint, as in the **347** DOWN_LINEAR phase. However, due to the actuators and the mechanics, the system has a response time to switch from the DOWN _UP **350** to the DOWN _LINEAR **347** speed. Therefore, during this phase, the actuators receive control signals to obtain a linear motion at the DOWN_LINEAR speed, but the pads have not reached that speed yet. At this stage, it is thus too early to re-clamp: the pads have to remain unclamped.
[0602] In a preferred embodiment, the DOWN_WAIT_CLAMP duration is fixed. In another preferred embodiment, the DOWN_WAIT_CLAMP travel range is fixed. In a third preferred embodiment, the duration is variable, and the state can be quit when the actual speed of the pads is close enough to the wanted speed. In a preferred embodiment, "enough" could be defined as the absolute value of the difference in percentage being lower than a predetermined threshold.
[0603] When the state is finished **430**, according to one of the criteria explained above, switch to DOWN_CLAMP **352**.
[0604] If the user translation speed setting becomes positive during this phase **402**, then switch to UP_UNCLAMP **343**. [0605] state **352** DOWN_CLAMP: during this state, the pads are re-clamped while maintaining the speed of the pads along the x axis equal to the user translation speed setpoint.
[0606] In a preferred embodiment, the DOWN_CLAMP duration is fixed. In another preferred embodiment, the DOWN_CLAMP travel range is fixed. In a third preferred embodiment, the duration is variable, and the state can be quit when the position of the pads along the y axis has reached a predetermined value. In a fourth preferred embodiment the duration is variable, and the state can be quit when the push force applied to the pads along the y axis has reached a predetermined value.
[0607] When the state is finished **410**, according to one of the criteria explained above, switch to DOWN_LINEAR **347**.

[0608] If the user translation speed setting becomes positive during this phase **402**, then switch to UP_UNCLAMP **343**.
[0609] An additional state is needed, which is not shown on FIG. **35** which is the initialization state. In a preferred embodiment, it will initialize all its variable and branch to the UP_LINEAR state **341**. In a preferred embodiment, the position of the left hand **15** along the x axis is at ¼ of the standard range, while the position of the right hand **16** is at ¾ of the standard range. The same applies to the z axis. Regarding the y axis, both hands are clamped.
[0610] When the translation speed setpoint (V.sub.T) changes, there is a jump from the UP_ to the DOWN_ states and vice-versa. If the value goes to 0, however, the state does not change. When V.sub.T=0, the state machine stops where it is, except for the UP_DOWN and DOWN_UP state, which continue until their end.
[0611] The invention has been described with reference to preferred embodiments. However, many variations are possible within the scope of the invention.

Claims

1. Catheter robot module for translation and rotation of a flexible elongated medical element, comprising: a casing, two pairs of movable pads: said pads of a same pair at least partly facing each other, each pair of movable pads being adapted to separately or in combination: perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle: clamping said flexible elongated medical element between said pads, translating forth said pads synchronously longitudinally in the same direction with respect to said casing, with respect to a user set longitudinal translation direction, unclamping said flexible elongated medical element, translating back said pads synchronously longitudinally in the same reverse direction with respect to said casing, with respect to said user set longitudinal translation direction, perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle: clamping said flexible elongated medical element between said pads, performing a relative forth translation of said pads transversely in opposite directions with respect to said casing, with respect to a transversal translation direction corresponding to a set rotation direction, unclamping said flexible elongated medical element, performing a relative back translation of said pads transversely in opposite reverse directions with respect to said casing, with respect to said transversal translation direction corresponding to said set rotation direction, a driver of said pairs of movable pads implemented so that: in at least one mode where, in combination, said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, and said rotation of said flexible elongated medical element is performed by at least one of said pairs of movable pads, conflict of synchronization between said translation and said rotation is managed at least: by varying travel extension of said forth translation in said first translation cycle for at least one of said pairs, and/or by varying travel extension and/or duration of said forth translation in said second rotation cycle for at least one of said pairs,   so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first translation cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second rotation cycle.
2. The catheter robot module according to claim 1, wherein: said driver of said pairs of movable pads is implemented so that: in one or several or all modes where said translation of said flexible elongated medical element is performed: said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled at least: by varying duration of said translating back in said first translation cycle for at least one of said pairs,   so as to control and keep said

phase opposition between both said pairs.

3. Catheter robot module for translation and rotation of a flexible elongated medical element, comprising: a casing, two pairs of movable pads: said pads of a same pair at least partly facing each other, each pair of movable pads being adapted to separately or in combination: perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle which clamps, performs a relative forth translation of said pads in opposite directions, unclamps, performs a relative back translation of said pads in opposite directions, depending on a set rotation direction, a driver of said pairs of movable pads implemented so that: conflict of synchronization, between said translation alternatively performed by said pairs of movable pads working in phase opposition and said rotation, when in combination, is managed at least: by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for at least one of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during said translation and during said rotation.

4. The catheter robot module according to claim 3, wherein: said driver of said pairs of movable pads is implemented so that: said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said phase opposition being controlled at least: by varying duration of said translating back in said first translation cycle for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.

5. The catheter robot module according to claim 1, preceding claims, wherein said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads.

6. The catheter robot module according to claim 5, wherein: said rotation of said flexible elongated medical element is alternatively performed by said pairs (**15**, **16**) of movable pads, both said pairs working in phase opposition, said phase opposition being controlled, at least: by varying duration of said translating back in said second rotation cycle for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.

7. Catheter robot module for translation and rotation of a flexible elongated medical element, comprising: a casing, two pairs of movable pads: said pads of a same pair at least partly facing each other, each pair of movable pads being adapted to separately or in combination: perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle: clamping said flexible elongated medical element between said pads, translating forth said pads synchronously longitudinally in the same direction with respect to said casing, with respect to a user set longitudinal translation direction, unclamping said flexible elongated medical element, translating back said pads synchronously longitudinally in the same reverse direction with respect to said casing, with respect to said user set longitudinal translation direction, perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle: clamping said flexible elongated medical element between said pads, performing a relative forth translation of said pads transversely in opposite directions with respect to said casing, with respect to a transversal translation direction corresponding to a set rotation direction, unclamping said flexible elongated medical element, performing a relative back translation of said pads transversely in opposite reverse directions with respect to said casing, with respect to said transversal translation direction corresponding to said set rotation direction, a driver of said pairs of movable pads implemented so that: said translation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said rotation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, conflict of synchronization between said translation and said rotation being managed, at least: by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for both of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first translation cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second rotation cycle.

8. The catheter robot module according to claim 7, wherein: said driver of said pairs of movable pads is implemented so that: said phase opposition is controlled, at least: by varying duration of said translating back, in said first translation cycle and in said second rotation cycle, for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.

9. Catheter robot module for translation and rotation of a flexible elongated medical element, comprising: a casing, two pairs of movable pads: said pads of a same pair at least partly facing each other, each pair of movable pads being adapted to separately or in combination: perform a translation of said flexible elongated medical element longitudinally with respect to said casing, by a first translation cycle which clamps, translates forth, unclamps, and translates back, depending on a user set longitudinal translation direction, perform a rotation of said flexible elongated medical element around longitudinal axis with respect to said casing, by a second rotation cycle which clamps, performs a relative forth translation of said pads in opposite directions, unclamps, performs a relative back translation of said pads in opposite directions, depending on a set rotation direction, a driver of said pairs of movable pads implemented so that: said translation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, both said pairs working in phase opposition, said rotation of said flexible elongated medical element being alternatively performed by said pairs of movable pads, conflict of synchronization between said translation and said rotation being managed, at least: by varying travel extension and/or duration of said forth translation, in said first translation cycle and/or in said second rotation cycle, for both of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during said translation and during said rotation.

10. The catheter robot module according to claim 9, wherein: said driver of said pairs of movable pads is implemented so that: said phase opposition is controlled, at least: by varying duration of said translating back, in said first translation cycle and in said second rotation cycle, for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.

11. The catheter robot module according to claim 1, wherein: said driver of said pairs of movable pads is implemented so that: said translation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, both pairs working in phase opposition, said phase opposition being controlled mainly or only: by varying duration of said translating back, in said first translation cycle for at least one of said pairs, so as to control and keep said phase opposition between both said pairs.

12. The catheter robot module according to claim 1, wherein: said driver of said pairs of movable pads is implemented so that: said rotation of said flexible elongated medical element is alternatively performed by said pairs of movable pads, conflict of synchronization between said translation and said rotation is managed mainly or only: by varying travel extension of said forth translation, in said first translation cycle for at least one of said pairs, so as to always keep at least one pair of movable pads clamped on said flexible elongated medical element, during the whole duration of said translation of said flexible elongated medical element in said first cycle as well as during the whole duration of said rotation of said flexible elongated medical element in said second cycle.

13. The catheter robot module according to claim 1, wherein said forth translation duration is always longer than said back translation duration.

14. The catheter robot module according to claim 1, wherein said varying travel extension of said forth translation in said first translation cycle for one of said pairs is performed by extending a predetermined standard forth translation travel range, reaching a value ranging from said predetermined standard forth translation travel range to a predetermined maximum forth translation travel range.

15. The catheter robot module according to claim 14, wherein said predetermined maximum forth translation travel range is comprised between 110% and 150% of said predetermined standard forth translation travel range, preferably between 120% and 140% of said predetermined standard forth translation travel range.

16. The catheter robot module according to claim 14, wherein said predetermined maximum forth translation travel range is split in two equal parts respectively at both ends of said predetermined standard forth translation travel range.

17. The catheter robot module according to claim 1, wherein there is some temporary overlapping between said flexible elongated medical element clamping by one of said pairs of movables pads and said flexible elongated medical element clamping by the other one of said pairs of movables pads, said temporary overlapping lasting preferably between 10% and 95% of the whole duration of said translation of said flexible elongated medical element.

18. The catheter robot module according to claim 1, wherein said flexible elongated medical element unclamping is performed simultaneously to a portion of said forth translation travel extension, during the second half of said forth translation travel extension, said portion ranging preferably from 5% to 20% of the full extent of said forth translation travel extension.

19. The catheter robot module according to claim 1, wherein said flexible elongated medical element clamping is performed simultaneously to a portion of said forth translation travel extension, during the first half of said forth translation travel extension, said portion ranging preferably from 5% to 20% of the full extent of said forth translation travel extension.

20. The catheter robot module according to claim 1, wherein said flexible elongated medical element clamping starts after the end of said back translation travel extension and after the beginning of next said forth translation travel extension.

21. The catheter robot module according to claim 1, wherein said varying duration of said translating back in said first translation cycle for one of said pairs, so as to control and keep said phase opposition between both said pairs, is performed by reducing or extending duration with respect to a standard back translation duration.

22. The catheter robot module according to claim 21, wherein said varying duration of said translating back in said first translation cycle for one of said pairs, so as to control and keep said phase opposition between both said pairs, is performed by reducing or extending duration with respect to a standard back translation duration less than requested for optimal phase opposition controlling and keeping so as to improve stability to the cost of higher number of cycles to get back at phase opposition target, a factor a of correction attenuation comprised between 0 and 1 being applied.

23. The catheter robot module according to claim 22, wherein said factor α of correction attenuation is comprised between 0.3 and 0.7, and is preferably about 0.5.

24. The catheter robot module according to claim 1, wherein said standard back translation duration is a decreasing function of a user command speed target value(s), for either translation and/or rotation, preferably minimum of both speed target values, when applicable, becoming selected user command speed target value.

25. The catheter robot module according to claim 24, wherein said decreasing function presents a central curved part which presents a concavity toward top and which is located between two horizontal parts.

26. The catheter robot module according to claim 25, wherein said central curved part is inversely proportional to said selected user command speed target value, whereas said horizontal parts are constant with respect to said selected user command speed target value.

27. The catheter robot module according to claim 1, wherein: said user set longitudinal translation direction can be varied continuously by said user, and/or said user set rotation direction can be varied continuously by said user.

28. The catheter robot module according to claim 1, wherein: said translation of said flexible elongated medical element longitudinally for first of said pairs of pads is performed by a first finite state machine, said rotation of said flexible elongated medical element around longitudinal axis with respect to said casing for first of said pairs of pads is performed by a second finite state machine, said translation of said flexible elongated medical element longitudinally for second of said pairs of pads is performed by a third finite state machine, said rotation of said flexible elongated medical element around longitudinal axis with respect to said casing for second of said pairs of pads is performed by a fourth finite state machine.

29. The catheter robot module according to claim 28, wherein: each of said finite state machines determines for a transition period between said forth translation and said back translation to go progressively from said forth translation to said back translation: start of said transition period, duration of said transition period, end of said transition period.

30. The catheter robot module according to claim 28, wherein each of said finite state machines changes periodically with a period which is less than 5 ms, preferably comprised between 0.5 ms and 2 ms, more preferably about 1 ms.